

Do Institutions Remove Inflation’s Grease?

Evidence from Brazil’s Indexed Minimum Wage*

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January 26, 2026

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Abstract

How do labor market institutions shape the propagation of inflation shocks? We address this question by studying Brazil’s annually indexed minimum wage in a high-inflation context. Conventional wisdom suggests that inflation can “grease” labor market adjustments, but institutional wage-setting may alter this mechanism. Using administrative data, we show that indexation creates upward nominal wage rigidity: workers exposed to the policy experience fewer month-to-month wage increases before indexation events, and firms anticipate the policy by rigidifying wages of workers who will be newly bound. We evaluate the macroeconomic implications by introducing a cost-push shock to a New Keynesian model with heterogeneous labor and an indexed minimum wage. While staggered indexation amplifies the inflation as grease mechanism by introducing nominal rigidities, anticipation dampens it via intertemporal substitution. Overall, amplification matters more since cost-push shocks have a stronger “greasing” effect compared to a setting where the minimum wage indexes every period. Our findings demonstrate that even in high-inflation environments, the institutional structure of wage-setting fundamentally shapes how shocks propagate through the economy.

*We are grateful to David Autor, Martin Beraja, Simon Jaeger, and Christian Wolf for their support and guidance at every stage of this project. We thank Constanza Abuin, Daron Acemoglu, Bautista Bassi, Pedro Martinez-Bruera, Tomas Caravello, Alberto Cavallo, Viola Corradini, Tishara Garg, Deivy Houeix, Pablo Ottonello, Martina Uccioli, Rafael Veiel, and Ivan Werning as well as participants of the MIT Labor and Macroeconomics lunches for their helpful comments and feedback.

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1 Introduction

Understanding the nature of nominal wage rigidity is a fundamental question in macroeconomics since it is one of the key mechanisms by which nominal shocks have real effects on output. For example, if nominal wages are rigid then an inflationary shock will lower the prevailing real wage and lead to an expansion of employment. This is popularly referred to as inflation “greasing the wheels” of the labor market (Tobin, 1972; Card and Hyslop, 1997), and is an argument often used in favor of targeting a positive inflation rate. However, to the extent that institutions determine nominal rigidities, and are themselves affected by inflation, they also play an important role in mediating the real effects of nominal shocks.

An important example of such an institution is the inflation-indexed minimum wage. This policy has become increasingly popular in recent years, with President Obama for example declaring in his 2013 State of the Union address that he would support indexing the federal minimum wage to inflation. To date, 19 states plus D.C. have indexed their own minimum wage to inflation.¹ In theory, such a policy would remove inflation’s grease altogether since it explicitly seeks to preserve the real minimum wage. However, in practice the indexation events do not take place in real-time with the inflation shocks, rather they do so on a pre-determined schedule. This creates stretches of time in between indexation events where nominal shocks may still have real effects.

In this paper, we study how Brazil’s indexed minimum wage affects inflation’s impact on low wage worker’s real wages. First, we find that workers exposed to the minimum wage are less likely to receive a nominal wage increase in between indexation events compared to workers who are never bound by the minimum wage. This implies that their real wages are more flexible downwards. Second, we show that wage setting anticipates the upcoming minimum wage increase. Workers who earn more than the current minimum wage but less than the upcoming one also receive fewer nominal wage increases throughout the year. To understand how the policy affects the propagation of nominal shocks in equilibrium, we introduce a cost-push shock to a New Keynesian model. The model is augmented with heterogeneous labor types and a time-dependent, backward-looking, inflation-indexed minimum wage. We show that the policy has opposing effects on inflation’s impact on the real wage. On the one hand, the nominal rigidities introduced by the policy amplify the inflation as grease mechanism. On the other hand, the policy introduces an anticipated marginal cost shock. This weakens the inflationary shock itself via the standard intertemporal substitution chan-

¹According to the [Economic Policy Institute](#), these are Alaska, Arizona, Arkansas, California, Colorado, Connecticut, Delaware, Florida, Hawaii, Illinois, Maine, Maryland, Massachusetts, Michigan, Minnesota, Missouri, Montana, Nebraska, Nevada, New Jersey, New Mexico, New York, Ohio, Oregon, Rhode Island, South Dakota, Vermont, Virginia, Washington, Washington D.C., and West Virginia.

nel, which indirectly dampens inflation’s ability to grease the wheels of the labor market. Finally, we show that under this policy the shock has a weaker long-run impact on inflation and output compared to a setting where the minimum wage indexes in every period.

We evaluate how Brazil’s inflation indexed minimum wage mediates the real effects of nominal shocks between January of 2015 and January of 2017. The policy takes the form of a national minimum wage that indexes once a year each January by the inflation accumulated over the past year. It is extremely salient since nearly 4% of all full time registered workers earn exactly the current minimum wage, while an additional 7% of them earn less than the following year’s minimum wage.² During this time period the country experienced one of the largest recessions in its history coupled with high inflation. Between 2015-2016, real gdp fell on average 3.4% per annum, and inflation averaged 8.9% per annum. In this context, the “inflation as grease” mechanism is particularly useful since it allows real wages to adjust downwards in response to the negative real shock without nominal wage cuts.

The empirical analysis focuses on the two indexation events in January of 2016 and January of 2017. For each of the two main results we first analyze the evidence in a non-parametric framework, and then in a regression framework. The analysis is based on monthly worker-level earnings observed in RAIS, an administrative employer-employee level dataset of all registered workers.

First, we present non-parametric evidence that workers bound by the minimum wage when they enter the sample are less likely to experience nominal wage increases throughout the year compared to workers not bound by it. This implies that their real wages are more flexible downwards. We measure upward nominal wage rigidity as the frequency of wage increases, which we proxy via sustained earnings increases. Between February and November, on average 12% of minimum wage workers experience a nominal wage increase in any given month, compared to 19% on average of non-minimum wage workers. This means that non-minimum wage workers are much more likely to receive a wage increase throughout the year. Naturally, this relationship flips in January when the minimum wage is indexed. 75% of minimum wage workers receive a wage increase compared to only 40% of non-minimum wage workers. The fraction of minimum wage workers who receive an increase in January is not 100% because we classify workers when they enter the sample so it is possible that they are above the threshold by the time the indexation event occurs.

Next, we find consistent evidence on differential wage rigidity across minimum wage exposure in a two-way fixed effects framework at the state-level. This strategy allows us to account for important confounders such as a worker’s relative position in the earnings

²An additional 60% of registered full time workers earn less than two multiples of the following year’s minimum wage.

distribution. Specifically, we measure variation in minimum wage bindingness at the state level using the Kaitz index. This index is defined as the distance between the current national minimum wage and the state-specific median of the earnings distribution. Then, we estimate how the frequency of wage increases varies at each earnings decile across states with different levels of minimum wage bindingness. First, we find that workers at the second decile of earnings, who are more likely to be exposed to the policy, are 9 percentage points less likely to receive a wage increase between February and November in states where the minimum wage is one standard deviation more binding. In January, they are 25 percentage points more likely to receive a wage increase. Higher earners, who are not exposed to the policy regardless of the state they live in, offer a placebo test. As expected, their wage rigidity exhibits almost no difference across more versus less exposed states. As a robustness check, we also estimate a worker-level version of the same regression where we control for additional individual-level characteristics such as age, gender, education, region, and sector. We find similar results.

Second, we show non-parametric evidence that wage setting anticipates the minimum wage increase. Here we focus on “non-binding minimum wage” workers, which we define as those who earn more than the current minimum wage but less than the upcoming minimum wage when they first enter the sample. They are informative because they are not bound by the mandated nominal floor. Instead, we would expect their wages to behave similarly to those of workers who are never exposed to the minimum wage. However, we find that non-binding minimum wage workers also experience more fewer nominal wage increases throughout the year relative to workers unexposed to the increase. On average, only 14% of them experience a wage increase each month. Since the only thing that differentiates them from workers who are just above the upcoming minimum wage is the fact that they lie below it, this fact is consistent with wage setting anticipating the upcoming increase and adjusting their wage setting behavior accordingly. While we remain agnostic as to why these workers do not experience wage increases on par with unexposed workers, a possible explanation could be the “inflation as conflict” hypothesis. Guerreiro et al. (2024) find that workers must take costly actions to ensure that nominal wages keep up with inflation. This is consistent with the idea that workers exposed to the minimum wage increase, who know that their wages will automatically catch up at the turn of the year, may be less willing to engage in these costly actions before then.

Next, we show that wage setting anticipates the upcoming minimum wage increase in an event study framework at the firm-level. This allows us to control for firm fixed-effects and other confounders such as a firm’s region, industry, and its serial exposure to the minimum wage. We measure firm-level exposure to the minimum wage as the share of workers who

earn less than the upcoming minimum wage, weighted by the distance between their average earnings in the base year and the value of the minimum wage in the following year. We find that the average earnings of workers at less exposed firms *rise* relative to more exposed firms, *before* the policy comes into effect. This downward-sloping pre-trend indicates that firms anticipate the policy. Unfortunately, it also means that we cannot interpret the results causally. Additionally, we test an alternative explanation for this downward sloping trend in the coefficients, namely that a firm is responding to past exposure, not anticipating future exposure, and that these two exposures are correlated. However, we control for past exposure and find no support for this hypothesis.

Motivated by these two facts, we incorporate this type of minimum wage policy into a New Keynesian model with two types of labor to study how a nominal shock will propagate. The model assumptions are informed by our empirical findings: low skill labor is paid exactly the nominal minimum wage which is rigid unless an indexation event occurs, and high skill labor is paid a completely flexible competitive wage. A representative household supplies both types of labor. The production side of the economy is composed of n competitive input firms, a continuum of monopolistically competitive intermediate firms, and a representative final good firm. The input firms hire high and low skill labor in different proportions, and sell their production at marginal cost to intermediate firms. Intermediate firms face nominal rigidities in price setting. This is the basis of an aggregate price Phillips curve to which the minimum wage is indexed. In the model, a period is defined at the quarterly level. The minimum wage updates once a year during the first quarter taking into account inflation over the previous four periods. This introduces a non-linearity in the model since the indexation rule is time-dependent.

In this setup, we study the impact of a cost-push shock. We focus on this type of shock to mimic the stagflationary environment in Brazil. In general, the variables respond as expected. Inflation rises, monetary policy reacts by setting a higher nominal interest rate, and output falls. However, as they return to steady state, their trajectory is interrupted by the first indexation event. With every new indexation event, inflation and the nominal interest rate spike upward while output spikes downwards, although by less than in response to the initial shock. The economy continues to cycle through this pattern following each subsequent minimum wage increase until the shock dissipates altogether and the nominal minimum wage stabilizes at a new higher level.

From this baseline scenario, we conclude that the timing of the shock matters. In the short-run, shocks that occur further from the indexation event have a stronger effect since inflation has more time to accumulate. However, they have the weakest effect in the long run. This is because inflation also has the most time to decline prior to the minimum wage

increase which means that it will rise from a lower base, and anticipation effects, which dampen the shock, are stronger.

Next, we consider two counterfactuals to evaluate the role of nominal rigidities and anticipation separately. In the first counterfactual, to understand the role of periodically adjusting the nominal minimum wage, we allow the minimum wage to index in every period. Under continuous indexation, the real wages of low skill workers fall by less relative to the baseline scenario. However, the increase in inflation and the decline in output are greater. Thus staggered indexation amplifies the inflation as grease mechanism.

In the second counterfactual, to understand the role of anticipation, we replace the cost-push shock to aggregate inflation with a cost-push news shock to the minimum wage. When the shock takes place, it raises the minimum wage which lowers output by raising marginal costs. When workers and firms learn about the upcoming shock, output falls in the present as well via intertemporal substitution, and thus generates deflation. Even though firms begin setting higher prices due to precautionary pricing as the indexation event approaches, it is not enough make inflation positive. In comparison to the baseline scenario, the real wage of low skill workers rises before the indexation event, and their employment falls by even more. Thus, the anticipation introduced by the policy indirectly removes inflation's grease.

Overall we find that a staggered indexation regime acts against a cost-push shock by introducing downward real wage flexibility and ultimately delivers a weaker response of inflation and output via the inflation as grease mechanism. The timing matters, with shocks that take place further away from the indexation event having a stronger short-run effect but a weaker long-run effect.

Literature: This paper contributes to the literature on how labor market institutions interact with shocks to determine real outcomes (Blanchard and Wolfers, 2000). More specifically, it speaks to the literature on nominal wage rigidities and how they mediate nominal shocks, and to the literatures on indexation and the minimum wage.

The literature that directly measures nominal wage rigidities has mostly focused on developed economies. Using a large representative survey in France, Le Bihan et al. (2012) find that the frequency of nominal wage adjustments is 38% at the quarterly frequency. Using the Survey of Income and Program Participation (SIPP) in the US, Barattieri et al. (2014) find a range between 21.1-26.6% for the probability that a US worker will experience a wage change in a given quarter. Finally, Sigurdsson and Sigurdardottir (2011) find that the frequency of monthly wage changes is 10.8% in Iceland. On average, these papers find that the monthly frequency of wage adjustment is approximately 10%. Our findings contribute to the literature on nominal wage rigidity by providing, to our knowledge, the first direct evidence of high-frequency micro-wage setting behavior in a large middle-income country. In

our context, the average monthly frequency of adjustments is approximately 20% throughout the year, and much higher at the turn of the year.

There is also a literature that studies the importance of nominal wage rigidities by evaluating the impact of monetary policy in differently rigid environments. Olivei and Tenreyro (2007) leverage variation in wage rigidity over time to make this point. They find that wage contracts in the US are more likely to adjust at the end of the fiscal year. Consistent with this fact, they find that monetary policy shocks have less impact on output when they take place right before this period of contract renewal and a larger one when they occur right after it. Similarly, Björklund et al. (2019) find that monetary policy shocks in Sweden have a 0.37 percentage point greater effect if they take place during periods where wage contracts are fixed relative to the average response. Minton and Wheaton (2022) leverage variation in wage rigidity across US states. They argue that states with a more binding minimum wage will have more rigid wages. Consistent with this, they find that employment responds more to monetary policy shocks in states with a relatively higher minimum wage. Finally, Faia and Pezone (2024) leverage variation in wage rigidity across firms in Germany. They find that firm's stock prices and employment respond more to monetary policy shocks in firms with more rigid wages as measured by their collective bargaining activity. Our paper relates most closely to Minton and Wheaton (2022) in that it leverages variation in wage rigidity arising from the minimum wage. However, we emphasize how the inflation-indexed minimum wage policy generates additional wage rigidity via an anticipation channel, and that its effects are stronger the greater is the inflationary shock. This indicates that the policy disproportionately exposes low-skill workers to the very shock it was designed to protect them from.

Seminal papers on wage indexation include Gray (1976) and Fischer (1977). They argue that indexation protects workers from nominal shocks but exposes them to real ones since it mechanically preserves the real wage. However, as highlighted by Jadresic (1996), this conclusion depends on the specific indexation rule used. Recent theoretical work has focused on determining what are the optimal indexation rules. Carrillo et al. (2022) find that workers prefer to index to past inflation when real shocks dominate, but they prefer to index to trend inflation when nominal shocks dominate. Recent empirical work has focused on evaluating indexation schemes in practice. For example, Manacorda (2004) finds that the *Scala Mobile* indexation scheme in Italy had a significant impact on wage inequality. Bijmans et al. (2023) finds that a temporary suspension of wage indexation in Belgium had positive employment effects in 2015. Our paper contributes to this literature by emphasizing that indexation events generally occur sporadically, and not continuously. Therefore, it is important to study how wage setting and other outcomes evolve in between and vis a vis indexation

events.

While economy-wide indexation is less prevalent now than in the late 20th century, the policy survives in many places in the form of partial indexation schemes. Jaeger et al. (2024) highlight the importance of two wage setting institutions that often embed indexation policies. These are collective bargaining agreements and the minimum wage. Koester and Grapow (2021), who provide a recent survey of the prevalence of different wage indexation policies in the euro area, find that 18% of euro area private sector workers work in countries where the minimum wage is indexed to inflation. The partial nature of these schemes allow researchers to leverage differential exposure to it within the same economy to identify its effects (Tito, 2011). Our paper fits into this tradition, and brings in evidence from a middle-income country context. We highlight the distributional impact of such policies, both for the workers that are directly affected by them and for firms that are differentially exposed to them.

Finally, our paper also contributes to the literature on the minimum wage. The vast majority of it focuses on the effect of unexpected and large increases in the minimum wage (Dube and Lindner, 2024). This is largely because it is easier to identify causal effects in this context. However, there is a broad swath of both local and national governments, such as California, France, and Canada, that have adopted or are moving towards adopting some form of inflation-indexed minimum wage policy (Karlmanangla, 2024; Campbell, 2024). Thus, it is important to study them, despite the identification challenges that arise from the foreseeable increases.

Outline: Our paper proceeds as follows. Section 2 provides background on the Brazilian minimum wage policy, describes the data, and presents descriptive evidence, including the non-parametric analysis. Section 3 analyzes how wage rigidity varies based on the bindingness of the minimum wage at the state- and worker-level in a two-way fixed effects framework. Section 4 presents the event studies that capture firm-level anticipation. Section 5 presents the model and discusses the implications of a backward-looking time-dependent inflation-indexed minimum wage for the propagation of inflationary shocks. Section 6 concludes.

2 Background, Data and Descriptive Evidence

2.1 The minimum wage in Brazil

Brazil has a long history of actively using minimum wage policy. There has been a nationwide minimum wage since 1984 which is closely watched because it also establishes the floor for pensions, welfare payments, and unemployment benefits. However, its use as an index

for other contracts is forbidden.

There are two other main wage-setting institutions in Brazil. While they are not explicitly indexed, the national minimum wage acts as a wage floor for both of them. These are (1) individual state's minimum wages, and (2) collective bargaining agreements. So far, five states set their own higher minimum wages.³ However, their application is irregular both in terms of who it applies to, and when they are updated. Collective bargaining agreements are negotiated and therefore respond endogeneously to the specific conditions of the sector or firm to which they apply. In contrast, the national minimum wage is updated on a regular schedule and applies to all full time workers. It does not take any individual sector or firm's characteristics into account.

Prior to 2008, the minimum wage was determined via negotiations between trade union centrals and the central government. As part of his platform, President Lula proposed that minimum wages should automatically adjust every January based on a rule that accounted for inflation over the previous 12 months and GDP growth with a two-year lag. This rule was signed into law in 2011 and used until 2019. A notable exception occurred in 2017 and 2018 when the two-year lag of GDP growth was negative. In these cases, the law was interpreted to mean that the minimum wage would only ever *increase* (and never decrease) based on these two factors. After 2019, the new government reverted to updating the minimum wage exclusively based on inflation. However, they maintained the regularity of updating it every January.

Every year at the beginning of September, the Brazilian government publishes its own forecast of the following year's minimum wage as part of its annual budget law (Lei Orçamentaria Anual, or LOA). This is necessary since so many social programs and other government payouts are tied to the minimum wage. While the minimum wage increase is already predictable since it relies on a fixed and transparent formula, this is yet another instance where the public is made aware of the expected upcoming increase.

The analysis period: January 2015-January 2017

Mainly for data access reasons, our analysis focuses on the period between January 2015 and January 2017. It encompasses two minimum wage hikes which we examine separately. In January of 2016, the monthly minimum wage for full-time workers rose from 788R\$ to 880R\$, implying an 11.67% increase. This was the result of an 11.27% inflation rate in 2015, and a 0.3% real GDP growth rate in 2014. The forecast published in the annual budget law proposal of September 2015 was 865R\$. In January of 2017 the minimum wage rose to 937R\$, a 6.5% increase from the year prior. It reflected the 6.6% inflation rate in 2016 but, as

³These are Rio de Janeiro, Rio Grande do Sul, Paraná, Sao Paulo and Santa Catarina.

explained above, it disregarded the -3.5% real GDP growth rate in 2015. The government’s forecast published in September 2016 as part of the annual budget law proposal was 945R\$.

2.2 Data on Earnings

We obtain monthly labor market data from RAIS (*Relação Anual de Informações Sociais*). This is an employer - employee matched annual mandatory survey for all registered firms in Brazil. From it, we create three different datasets. Section 3 uses a worker-level and a state-level dataset, while section 4 relies on a firm-level dataset. In all samples, we restrict to private sector workers with a standard 44 hour workweek aged 18 to 54, employed by firms with more than 5 workers. The final sample contains approximately 4 million workers per year, in roughly 300,000 firms, spread across 27 states. While these sample restrictions are restrictive, recent work by Derenoncourt et al. (2021) shows that the minimum wage has a strong pass-through into the informal sector. In future work we plan to explicitly account for how and to what extent our results carry over to Brazil’s large low-wage, informal, and small-firm sector. For now, we rely on existing work to hypothesize that they will.

In addition to a worker’s characteristics such as age, gender, race and education, we observe their earnings in two ways. First, firms directly report a worker’s annual average monthly earnings. For worker l in year t , we label this variable $\hat{y}_t^l$. Second, since 2015, firms also report worker’s monthly earnings in each month. For worker l in month m and year t we label this variable as y_{mt}^l . Firms are required to report how much they effectively pay each worker, including overtime but excluding the 13th salary.⁴

In our dataset, sometimes a worker’s directly reported annual average monthly earnings $\hat{y}_t^l$ does not match the annual average predicted by averaging monthly earnings y_{mt}^l . This is likely due to measurement error in the monthly earnings, not in the directly reported annual average, since it is the latter that dictates eligibility for welfare benefits. See appendix A.1 for a fuller discussion. To address this issue, we restrict the sample to workers whose directly reported annual average monthly earnings are consistent with their monthly earnings.

Wage rigidity. In the analysis, we measure how nominal wage rigidity varies with exposure to the minimum wage. This requires us to measure wage changes at the worker level. To proxy for wage changes, we rely on sustained positive changes in earnings. This is a suitable proxy for two reasons. First, sustained earnings changes are less likely to reflect transitory changes in earnings due to factors such as changes in hours worked, or one-off bonus payments. Second, it is illegal for firms to unilaterally lower a worker’s nominal

⁴The 13th salary is an annual bonus that is paid out in two parts. The first half must be paid before November, and the second half must be paid before December.

wages.⁵ We define a sustained earnings increase as a month-to-month increase in earnings that is at least maintained the following month regardless of whether a worker switches employers.

Summary Statistics. To get a better understanding of the characteristics of workers exposed and unexposed to the minimum wage, Table 1 describes workers along the wage distribution for the 2015-2016 sample.⁶ We use their 2015 annual average monthly earnings $\hat{y}_t^l$, to sort them. Each column corresponds to one of three groups: those earning less than the following year’s minimum wage, those earning between the following minimum wage and less than two multiples of the following year’s minimum wage, and those earning more than that.

Workers who earn less than the following year’s minimum wage are surprisingly similar to those who earn up to two times the minimum wage. They are more likely to work in services, have a high school degree or less, and work in smaller establishments. In contrast, workers who earn more than two times the minimum wage are more likely to work in manufacturing, have a college degree, and work in larger establishments. Overall, as earnings increase the likelihood of living in the northeast decreases, and that of living in the southeast increases.

2.3 Non-parametric analysis

We start with a non-parametric analysis of wage-setting at the worker level. The objective is twofold. First, to show that the nominal wages of minimum wage workers are less likely to rise compared to non-minimum wage workers, and second, to show that firms anticipate the upcoming increase.

We measure wage-setting using the monthly frequency of wage increases, defined as the share of workers who experience a sustained earnings increase in a given month. For example, if 20% of workers experience a sustained earnings increase in each month and these events occur independently over time, this implies an average duration of five months between wage increases. A higher frequency of nominal wage increases implies that the real wage is more rigid downwards, while a lower frequency implies that it is more flexible downwards.

Figure 1 plots the frequency of sustained earnings increases for four types of workers at the monthly level between February and November. Each worker-type has varying degrees of exposure to the minimum wage. To capture possible anticipation, we focus on the period before each minimum wage increase. Panel 1a reports the results for 2015 corresponding to the January 2016 minimum wage increase, while panel 1b reports the results for 2016,

⁵See Article 7, item VI of the 1988 constitution, and article 468 of the labor code (*Consolidação das Leis do Trabalho, CLT*).

⁶The patterns are similar for the 2016-2017 sample, see appendix A.2.

Table 1: Summary Statistics by Earnings Percentiles (2015)

	$\hat{y}_t < mw^{t+1}$	$mw^{t+1} < \hat{y}_t < 2 \cdot mw^{t+1}$	$2 \cdot mw^{t+1} < \hat{y}_t$
Observations	1000101	6288112	3068607
Female	0.45	0.40	0.23
Age	32.19	34.07	37.80
Education			
Less than High School	0.53	0.43	0.29
High School	0.45	0.52	0.45
More than High School	0.02	0.05	0.27
Establishment Size			
Small (<20)	0.19	0.22	0.16
Medium (20–250)	0.37	0.39	0.41
Large (>250)	0.44	0.39	0.44
Regions			
North	0.08	0.05	0.05
Northeast	0.37	0.16	0.10
Southeast	0.37	0.51	0.57
South	0.08	0.19	0.20
Centerwest	0.09	0.08	0.08
Sectors			
Agr. and Mining	0.17	0.08	0.05
Manuf. and Constr.	0.30	0.31	0.41
Services	0.53	0.60	0.52

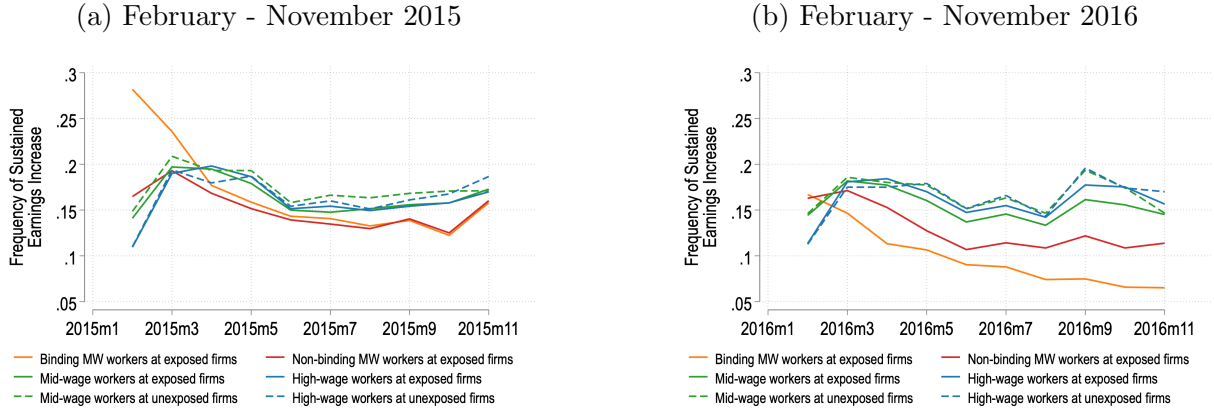
Notes: This table presents summary statistics for workers in three groups defined by their average monthly earnings in 2015 relative to the upcoming minimum wage increase in January 2016. The first column includes workers who earn less than the 2016 minimum wage, the second column groups workers who earn more than that but less than two multiples of it, and the third column includes workers who earn more than two multiples of it.

corresponding to the January 2017 minimum wage increase.

Workers are classified into one of four types based on the first observation of their monthly nominal earnings. “Binding MW” and “Non-binding MW” workers are both exposed to the minimum wage increase in the sense that they earn less than the following year’s minimum wage. The difference between them is that binding MW workers (orange line) earn exactly the current minimum wage when they enter the sample while non-binding MW workers (red line) earn more than it.

“Mid-wage” and “High-wage” workers are both unexposed to the minimum wage increase in the sense that they earn more than the following year’s minimum wage. Mid-wage workers (green lines) earn less than two multiples of next year’s minimum wage while high-wage workers (blue lines) earn more than that. These unexposed worker types are additionally sorted into one of two firm types: exposed or unexposed. Exposed firms (solid lines) employ at least one exposed worker, while unexposed firms (dashed lines) employ none.

Figure 1: Non-parametric Evidence on Upward Nominal Wage Rigidity



Notes: This figure presents non-parametric evidence on the frequency of sustained earnings increases at the worker-level for the frequency of nominal wage increases. The frequency of sustained earnings increases is defined as the share of workers with a month-to-month earnings increase that is at least maintained the following month. Workers are classified when they enter the sample. Binding MW workers are those that earn exactly the current minimum wage. Non-binding MW workers are those that earn more than the current minimum wage but less than the following year’s minimum wage. Mid-wage workers are those that earn more than the following year’s minimum wage but less than two multiples of it. High-wage workers are those that earn more than two multiples of the following year’s minimum wage.

Non-parametric evidence on differential wage rigidity:

Figure 1 shows that exposed workers experience fewer nominal wage increases throughout the year than unexposed workers. This is true in both sample periods. In general, binding MW and non-binding MW workers experience a frequency of sustained earnings increases

of around 12-14% throughout most of the year, while mid-wage and high-wage workers experience a frequency of around 18-20%. While not shown in the figure, this relationship flips at the turn of the year when the indexation event takes place. Exposed workers are twice as likely to experience a sustained earnings increase, 75% of them do, compared to unexposed workers, only 40% of them do.

To make progress on what may explain the correlation between minimum wage exposure and wage rigidity, we compare wage setting behavior across worker and firm types.

We find little evidence that it is the worker’s level of earnings that drives the results. To support this claim, we compare the wage setting behavior of mid-wage relative to high-wage workers. While these two groups have significantly different earnings levels, they exhibit a strikingly similar frequency of earnings changes. This is especially meaningful given that these two types of workers are very different in terms of their characteristics, as reported in Table 1.

Finally, we find little support for the hypothesis that firm type drives the results. For this, we contrast similar workers in differently exposed firms. We find that mid-wage and high-wage workers at both exposed and unexposed firms have a similar wage rigidity. This suggests that other firm characteristics that may be correlated with firm-level minimum wage exposure, such as a firm’s region or sector, are unlikely to drive the result. Appendix A.3 extends this exercise to consider a finer categorization of firms and finds similar results, with some caveats. These findings further support the hypothesis that worker-level exposure is the most important determinant of earnings rigidity.

Non-parametric evidence on anticipation:

The behavior of the non-binding MW workers suggests that firms anticipate the upcoming minimum wage increase. This group of workers is informative because they earn more than the prevailing minimum wage. Thus, it cannot be that the rigidity of their earnings is explained by the policy’s mechanical bindingness. Instead, what separates them from unexposed workers and ties them to binding MW workers, is that they will be affected by the upcoming minimum wage increase. Hence, any similarities in their wage setting behavior can be attributed to this future exposure. Although we remain agnostic as to why exposure lends wage-setting power to firms, the empirical patterns are consistent with this being true. A possible explanation put forward by Guerreiro et al. (2024) suggests that workers must take costly actions to ensure that nominal wages keep up with inflation. Minimum wage workers, who know that their wages will catch up automatically at the turn of the year, may be less willing to engage in these actions before then.

3 Wage Rigidity varies under the Minimum Wage

The first empirical result, for which we provided non-parametric evidence, is that workers exposed to the minimum wage increase experience fewer nominal wage increases throughout the year. This implies that their real wages are more flexible downwards during this period. However, exposure to the minimum wage may be correlated with wage setting practices for reasons other than the policy itself. To explore this possibility, we estimate a regression version of the non-parametric analysis that leverages variation in minimum wage bindingness across states to better address possible unobserved confounders.

The biggest concern is that differential wage-setting across exposed and unexposed workers is due to a worker’s relative position in the earnings distribution and not their exposure to the minimum wage. These two characteristics are correlated since it is low-wage workers that are exposed to the policy. The comparison of mid-wage and high-wage workers in the non-parametric analysis begins to address this concern by showing that the earnings rigidity of unexposed workers is similar at different points in the earnings distribution. However, it is not conclusive because there may be a non-linear relationship between a worker’s position in the earnings distribution and how their wages are set.

To directly address this concern we compare wage-setting within the same earnings decile across states where the minimum wage is more versus less binding. This allows us to isolate the effect of minimum wage bindingness from a worker’s relative position in the earnings distribution. The results are consistent with the non-parametric evidence. Throughout the year, we find that the frequency of wage adjustment for workers at the 2nd decile of earnings is 9pp lower in areas where the minimum wage is one standard deviation more binding. At the turn of the year, they are 25pp more likely to experience a wage increase. As a robustness check, we estimate a worker-level version of the same regression. This allows us to control for additional individual-level characteristics such as age, gender, education, region, and sector. We find similar results.

3.1 State-level Analysis

Econometric Framework

To correlate the minimum wage with the frequency of sustained earnings increases, we estimate a state-level two-way fixed effects model relying on state-level variation in minimum wage exposure. Since figure 1 shows that wage-setting behaves differently throughout the year versus at the turn of the year, we estimate the model separately for the period between February to November and the period from December to January. We pool the data prior

to each minimum wage increase from the 2015-2016 and 2016-2017 event-windows. In the regression, the event-window is indexed by t . So, for example, January of 2016 belongs to the 2015-2016 event-window, while February of 2016 belongs to the 2016-2017 event-window.

We measure state-level minimum wage exposure using the Kaitz index (Lee, 1999; Autor et al., 2016). The Kaitz index in state s is defined as the difference between the log national minimum wage and the log state-median wage: $Kaitz_s = \log(mw) - \log(\hat{y}_s^{median})$. The measure captures how binding the minimum wage is in each state.

We specify the model as follows:

$$F_{smt}^d = \sum_{k=1}^{10} \beta^d (\mathbb{1}_{d=k} \times Kaitz_{st}) + \rho_d + \gamma_{sd} + \delta_{td} + \epsilon_{smt}, \quad (1)$$

Where F_{smt}^d is the frequency of sustained earnings increases in state s at month m -event window t for earnings decile d . ρ_d are earnings decile fixed effects. γ_{sd} are state-by-decile fixed effects and δ_{td} are event-window-by-decile specific fixed effects. The sample includes all workers in firms with more than 5 employees. In each sample, workers are categorized into each earnings decile based on their first observed monthly earnings. Standard errors are clustered at the state level.

The state-by-decile fixed effects control for unobserved state-level characteristics within each decile that are fixed over time. An important example is sectoral composition. If lower earnings deciles in poorer states also have a greater share of their economic activity in a specific sector, such as agriculture, then wage-setting practices specific to that economic activity will correlate with the minimum wage. However, since we do not expect aggregate sectoral composition within an earnings decile to change significantly over the short two-year period under consideration, the fixed effects will account for this correlation.

The event-window-by-decile fixed effects account for aggregate shocks that are common across states but may correlate with the bindingness of the minimum wage over time within each decile. The minimum wage is constant within each event-window and therefore so is its bindingness. Thus, only event-window specific fixed effects are necessary for this purpose. An important example of an aggregate shock this accounts for is the inflationary environment. Since inflation mechanically leads to a higher minimum wage, it is also associated with greater bindingness. The fixed effects account for any correlation between such aggregate shocks and wage setting practices.

This design allows us to compare workers in the same position of the earnings distribution in states where the minimum wage is more versus less binding. The coefficients of interest are the β^d coefficients which isolate the effect of minimum wage bindingness accounting for

permanent differences within each decile across states, and within each decile across event windows. Higher earnings deciles are a natural placebo group for lower earnings deciles since the minimum wage is less likely to matter for them, regardless of the state-specific minimum wage bindingness.

Engbom and Moser (2022), who study the correlation between the minimum wage and wage inequality, also employ an empirical strategy that relies on the Kaitz index in the context of Brazil. They do two things differently. First, they define the Kaitz index around the 90th percentile of earnings instead of the median. This is because they find that the minimum wage has positive and significant spillovers up to the 90th percentile of the earnings distribution. Second, they include a quadratic term of the Kaitz index to capture non-linear effects of minimum wage bindingness. Our results are robust to these modifications.

State-level Results on Differential Wage Rigidity

Figure 2 plots the ten β^d coefficients from estimating equation 1 for each set of months. Panel 2a pools the months between February and November, while panel 2b pools together December and January. The standard deviation of the Kaitz index is 0.1. Thus, $\beta^d \times 0.1$ describes the percentage point change in the frequency of sustained earnings increases when the minimum wage bindingness increases by one standard deviation.

Panel 2a shows that workers at the bottom of the wage distribution are less likely to experience a monthly wage increase in states where the minimum wage binds more. The negative and significant β^2 coefficient implies that a one standard deviation increase in the Kaitz index is associated with a 9 percentage point decrease in the frequency of sustained earnings increases for workers at the 2nd decile of the earnings distribution. This effect is economically significant. Given that the average frequency of sustained earnings increases for these workers is 12.6%, this implies a 71% decrease relative to the mean.

Panel 2b shows that this relationship flips at the turn of the year. Workers at the bottom of the earnings distribution have more flexible earnings in states where the minimum wage binds more. The positive and significant β^2 coefficient implies that a one standard deviation increase in the Kaitz index is associated with a 25 percentage point increase in the frequency of sustained earnings increases for workers at the 2nd earnings decile. Given that the average frequency of sustained earnings increases for these workers is 61.3%, this implies a 40% increase relative to the mean.

In both panels, workers higher up in the earnings distribution represent a placebo test. Since they are not exposed to the minimum wage regardless of their state’s minimum wage bindingness, they should behave similarly across all locations. Consistent with this fact, for higher earners the effect on the frequency of wage increases of being in a state with a more

binding minimum wage is not significantly different from zero.

Overall, both the direction and magnitude of the results for the lower earnings deciles and the higher earnings deciles are consistent with the non-parametric evidence presented in figure 1. They provide further evidence that nominal wage setting is more rigid for workers exposed to the minimum wage between February and November, but more flexible at the turn of the year.

3.2 Robustness: Worker-level Analysis

Econometric Framework

To further address concerns about unobserved confounders, we estimate a worker-level version of equation 1. This allows us flexibly to control for other worker-specific characteristics such as gender and age in addition to their earnings decile. We proceed by regressing a dummy variable indicating whether a worker l experiences a sustained earnings increase on the interaction of the Kaitz index with the worker’s state-specific earnings decile.

$$D_{lsm}^F = \sum_{k=1}^{10} \lambda^d (\mathbb{1}_{d=k} \times Kaitz_{st}) + \rho_d + \gamma_{sd} + \delta_{td} + X_l' + \epsilon_{lsm}, \quad (2)$$

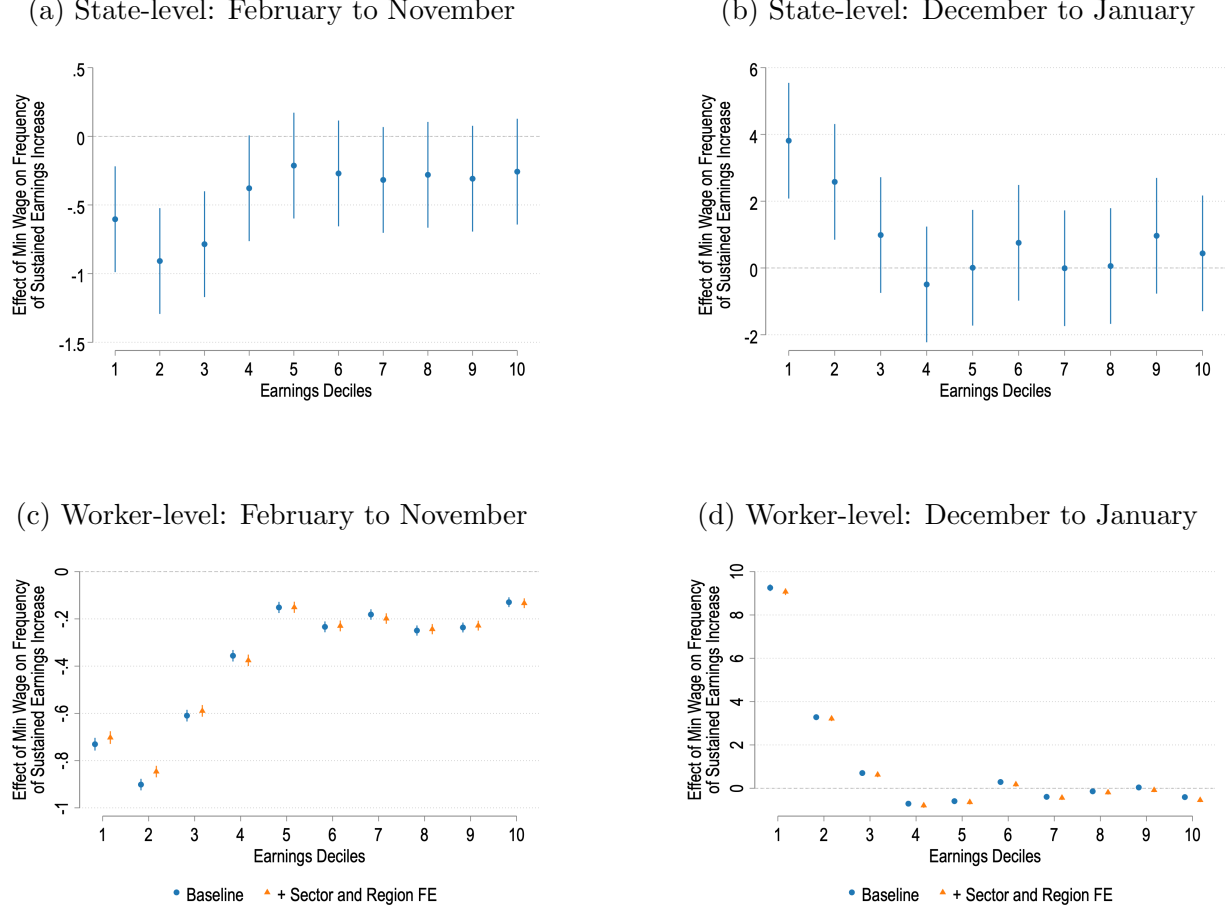
Where D_{lsm}^F is a dummy variable that equals one if worker l in state s at month m event-window t experiences a sustained earnings increase. As before, ρ_d are earnings decile fixed effects, γ_{sd} are state-by-decile fixed effects and δ_{td} are event-window-by-decile fixed effects. X_l' is a vector of worker-specific controls. This includes age, education, gender, sector, and region. The sample includes all workers in firms with more than 5 employees. Standard errors are clustered at the state level.

The coefficient of interest is λ^d . It describes how the likelihood of a wage increase differs in each earnings decile d in states where the minimum wage is more versus less binding.

Worker-level Results on Differential Wage Rigidity

Figures 2c and 2d plot the λ^d coefficients from estimating equation 2 and respectively pooling together the months from February to November and December to January. The blue circles represent the coefficients from a baseline regression that includes controls for age, education, and gender. The orange triangles represent the coefficients from a regression that additionally includes sector and region fixed effects. The results are practically unchanged between the two specifications.

Figure 2: State-level Regressions



Notes: Panels 2a and 2b figure plots the β^d coefficients from estimating equation 1 of the frequency of sustained earnings increases on the Kaitz index at each earnings decile, controlling for state-decile and time-decile fixed effects. The frequency of sustained earnings increases is defined as the share of workers with a month-to-month earnings increase that is at least maintained the following month. Standard errors clustered at the state level. Panels 2c and 2d plot the results from a single worker-level regression. The blue circles represent the λ^d coefficients described by equation 2. Specifically, the coefficients describe the change in the probability that a worker experiences a sustained earnings increase at each earnings decile as the state-level bindingness of the minimum wage increases. We control for worker characteristics such as age, education, gender and earnings deciles. The orange triangles represent the same λ^d coefficients from a regression that additionally controls for industry and region fixed effects.

The worker-level results are consistent with the state-level analysis. Workers at the 2nd earnings decile in states where the minimum wage binds more are less likely to experience a wage increase. The point estimates imply that a 2nd earnings decile worker in a state where the minimum wage binds by 1 standard deviation more, is 9pp less likely to experience a wage increase throughout the year and 32pp more likely to experience it at the turn of the year.

4 Firms Anticipate the Minimum Wage

The second empirical result, for which we provided non-parametric evidence, is that wage setting anticipates the minimum wage increase. In this section we build on this evidence using an event study framework. It allows us to evaluate how wage setting varies in firms that are more versus less exposed to the minimum wage increase, before and after it occurs. By focusing on the period before the indexation event, we can capture any anticipation.

We find that, leading up to the minimum wage increase, the average earnings of workers at less exposed firms increase relative to those at more exposed firms. In the figures, this presents as a downward slope in the coefficients prior to the indexation event. We also consider two important robustness checks. First, we show that the results are not driven by region or sector specific trends that may be correlated with exposure. Second, we show that they are also not driven by serial correlation in exposure.

We start by describing the firm-level measure of exposure to the minimum wage which we denote Z_f . Then, we describe the empirical specification that we use to evaluate its impact during the two 24-month event windows spanning from January 2015 to December 2016, which covers the January 2016 increase, and January 2016 to December 2017, which covers the January 2017 increase. Finally, we discuss the baseline results and robustness checks.

The Firm-level Exposure Measure

The exposure measure Z_f captures the predicted mechanical average earnings change for workers at firm f due to the minimum wage increase. It is the analog to what Harasztsi and Lindner (2019) call the Wage Gap measure. If there is full compliance with the policy, no spillovers, and no disemployment effects, or if these exactly cancel each other out, it will exactly predict the actual change in the average earnings of workers at firm f in January.

Equation 3 describes how Z_f is calculated. It averages the expected earnings increase of exposed and unexposed workers when the minimum wage indexes. Exposed workers are those whose annual average monthly earnings fall below the following year’s new minimum wage. Since the minimum wage always updates in January, we use the calendar annual average in

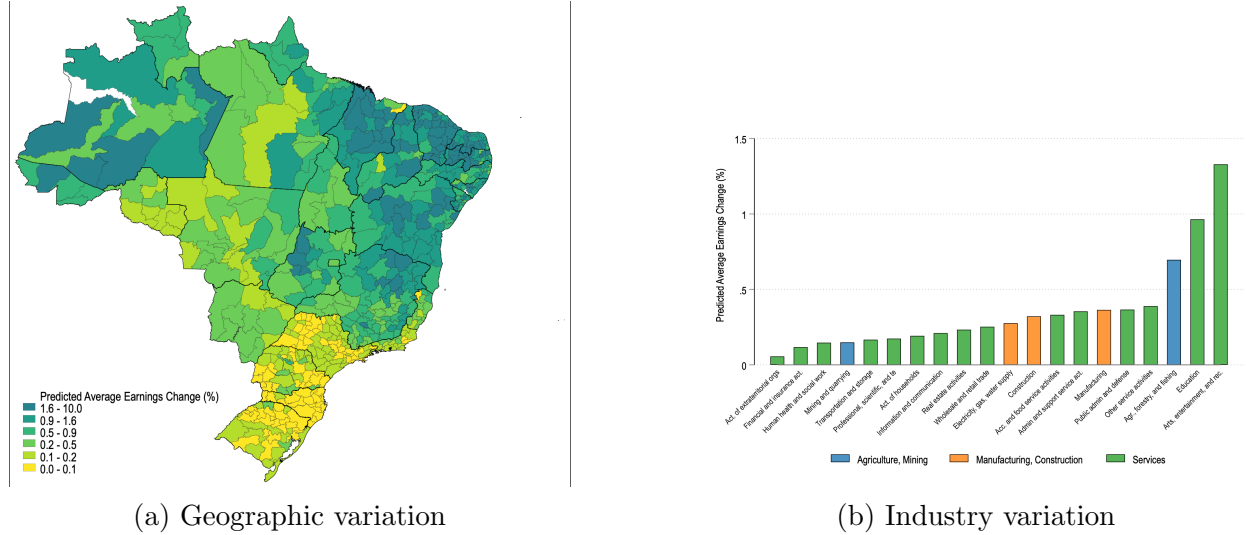
the previous year to avoid endogeneity concerns arising from earnings fluctuations in any specific month (especially December). The max operator takes on the value of zero for unexposed workers. Thus, the instrument incorporates information both on the extensive margin of exposure, how many workers at a firm are exposed, and the intensive margin, by how much.

$$Z_f = \frac{1}{N_f} \times \sum_{l \in N_f} \max\{mw_{t+1} - \hat{y}_{ft}^l, 0\}, \quad (3)$$

N_f is the total number of workers employed in firm f in December right before the minimum wage increase. mw_{t+1} is the new log minimum wage the following January. $\hat{y}_{ft}^l$ is the log directly reported annual average monthly earnings of worker l in firm f and year t who is employed in December.

We calculate Z_f separately for the two event windows. Both in the 2015-2016 and 2016-2017 samples, approximately 20% of firms have a positive value of treatment. For them, the average value of exposure is 1.6%.

Figure 3: Exposure to the minimum wage in Brazil



Notes: The figures show the average variation in the exposure measure Z_f from 2015-2016 separately at the geographic and industrial levels. Panel 3a shows the average value of the treatment in each of the 558 microregions. Panel 3b shows the average value of the treatment across each of the 20 industries.

Figure 3 shows how the instrument varies across regions and industries. Panel 3a shows the average value of the instrument at the microregion level between 2015-2016. On average, the northeast region of the country is much more exposed (darker colors) than the south-

east (lighter colors). This is consistent with the fact that the southeastern region is more economically developed.

Panel 3b shows the average value of treatment at the broad industry level between 2015-2016. On average, the service sector is very exposed to the minimum wage, especially in cases like the accommodation and food services industry, or the wholesale and retail trade industry. The agricultural sector is also relatively highly exposed. As might be expected, the least exposed industries are the financial and insurance activity industry, and the mining and quarrying sector.

Empirical Framework

A firm-level event-study model, described in equation 4, isolates how exposure correlates with the minimum wage over time in each 24-month event window. Since the minimum wage increase is predictable, we cannot interpret these coefficients causally. We interact the exposure measure Z_f with month-by-year dummies $\mathbb{1}_{m=k}$, and include firm γ_f and month-by-year δ_m fixed effects separately. The period $m = 0$ corresponds to the month where the minimum wage increases. In the 2015-2016 sample this is January 2016, and in the 2016-2017 sample this is January 2017. So, we normalize the $\beta_{m=-1}^{Firm}$ coefficient to 0. The coefficients of interest β_m^{Firm} capture the change in outcome y_{fm} in more versus less exposed firms in each month relative to their baseline difference in December of the pre-period.

$$y_{fm} = \sum_{\substack{k=11 \\ k \neq -1 \\ k=-12}} \beta_m^{Firm} (\mathbb{1}_{m=k} \times Z_f) + \gamma_f + \delta_m + \varepsilon_{fm}, \quad (4)$$

When the outcome is log average earnings, the benchmark value of the coefficient β_0^{Firm} is one. This happens when the exposure measure Z_f perfectly predicts the actual change in average earnings. It is also important to note that the nature of the instrument's construction normalizes the expected effect of the minimum wage. For example, whether the increase is large or small, whether a unit is very exposed or not, the baseline expected effect on log average earnings is one. In this sense, the coefficient is more comparable to an elasticity than to an absolute effect. While it can be informative of the underlying mechanisms, it may mask the actual impact of the minimum wage in reality. To obtain a sense of the actual impact, we must account for the underlying value of the instrument itself.

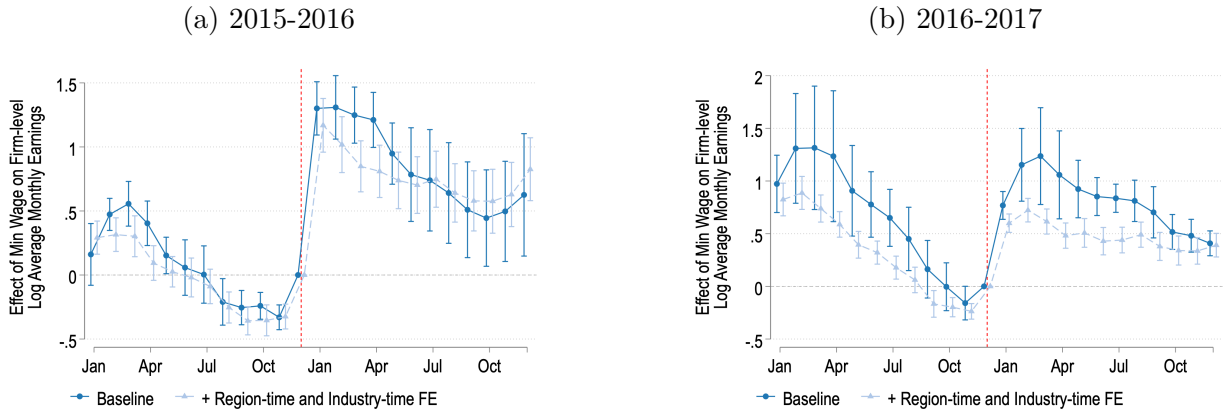
To account for alternative explanations of the trends observed in the baseline results, we consider two robustness checks. In the first, we account for the possibility that region or sector specific trends correlate with minimum wage exposure. For example, if agricultural

workers are more likely to be exposed to the minimum wage and their wage setting practices also happen to align with the minimum indexation because of the timing of harvests, we could be misattributing the results to anticipation. To address this, we re-estimate the baseline model and include region-by-month-year and sector-by-month-year fixed effects. In the second robustness check, we account for the possibility that serial correlation in exposure explains the results. It could be that the coefficients are reflecting a firm's reaction to past shocks, and that these are correlated with future ones. To explore this, we re-estimate the baseline model and include the one, two and three year lag of the exposure measure by interacting it with month-by-year dummies.

Firm-level Results on Anticipation

Figure 4 plots in dark blue circles the baseline β_m^{Firm} coefficients from estimating equation 4 using log average earnings as an outcome for each of the two event windows separately. Panel 4a plots the results for 2015-2016 and Panel 4b plots the results for 2016-2017. In the results presented here, we weight the regressions by the average number of workers in each firm.

Figure 4: Earnings at the Firm level



Notes: The figure plots the event study coefficients β_m^{Firm} from estimating equation 4 using firm-level average earnings as the outcome. In dark blue circles we report the coefficients from the baseline regression, and in light blue triangles we report the coefficients from the regression controlling for firm's region and sector. The regressions are weighted by each firm's average employment during the sample period.

The first takeaway is that in January log average earnings increase significantly in more exposed versus less exposed firms relative to their baseline difference in December prior. This suggests that our exposure measure captures some relevant aspect of how the minimum wage policy correlates with firm wage setting. This difference remains positive and significant throughout the year. Also, the patterns are quite similar across the two samples. This

suggests that the effects are not coincidental, or explained by another concurrent event, but rather a feature of wage setting under the minimum wage policy.

Second, prior to the indexation event the estimates are significant and they slope downward. This is consistent with anticipation effects. The trend implies that log average earnings at less exposed firms are increasing compared to exposed firms and relative to their difference in the base period. This is consistent with a story where unexposed workers are more likely to receive earnings increases throughout the year relative to exposed workers.

Robustness: Region and sector trends do not explain the downward trend.

The light blue triangles in figure 4 present the results from a robustness check where we control for region-by-month-year and sector-by-month-year fixed effects in addition to the firm and month-year fixed effects included in the baseline specification. The results are very similar to the baseline, suggesting that other firm characteristics which may be correlated with exposure do not drive our findings.

Robustness: Past exposure does not explain the downward trend.

Panels 5a and 5c report, in teal triangles, the results from a robustness check where we control for lags of the exposure measure. They also report the baseline coefficients in dark blue circles for comparison. The results show that there is almost no difference between the two specifications. In both, the downward trend is present, and average log earnings respond strongly and positively to the indexation event.

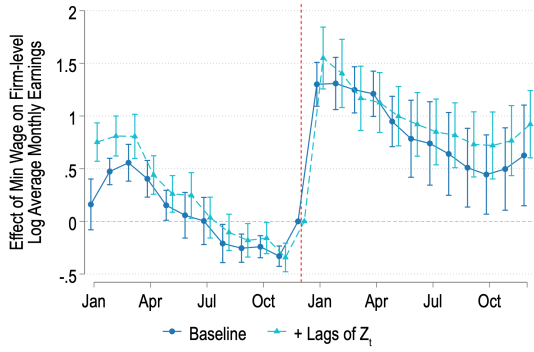
Panels 5b and 5d report the coefficients on the lagged exposure values themselves from the same regressions reported in the other two panels. While lagged exposure has some predictive power, it is much smaller in magnitude than the current exposure variable, and doesn't do so in a consistent manner. These results support the view that the downward slope prior to the event is more likely due to anticipation than to serial correlation in exposure.

5 Model

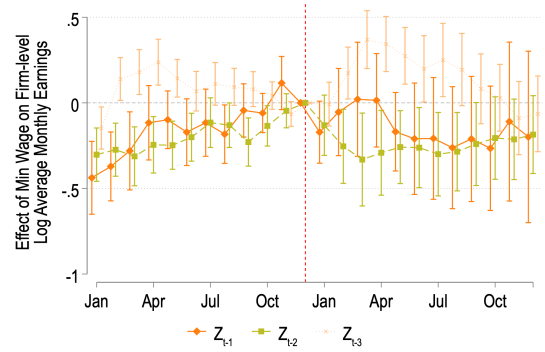
In this section, we present a New Keynesian model augmented with two types of labor, high and low skill, and an inflation-indexed minimum wage that updates once every four quarters. The model assumptions are informed by our empirical findings. First, the rigidity of the nominal minimum wage in between indexation events mimicks the empirical fact that minimum wage workers are less likely to receive a nominal wage increase throughout the year. Second, anticipation is possible since agents have full information about the indexation rule

Figure 5: Serial correlation does not explain downward pre-trend

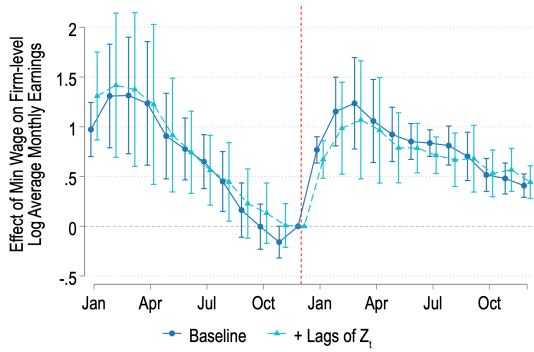
(a) 2015-2016: Effect of Current Exposure



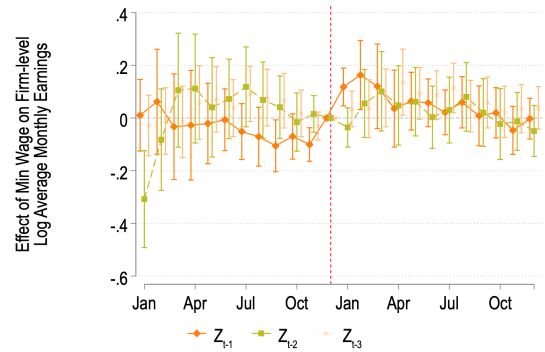
(b) 2015-2016: Effect of Past Exposure



(c) 2016-2017: Effect of Current Exposure



(d) 2016-2017: Effect of Past Exposure



Notes: The figure plots the coefficients from estimating equation 4 using log average monthly earnings as the outcome, and controlling for the 1-, 2-, and 3-year lag of treatment. Panels 5a and 5c report the event study coefficients β_m^{Firm} showing the effect of the treatment, while panels 5b and 5d report the effect of the lags themselves. The regressions are weighted by the average employment in each firm.

and the occurrence of inflation. Thus, they know exactly when and by how much the nominal minimum wage will increase.

In the baseline scenario, we introduce an inflationary shock in the form of a cost-push shock. Then, to evaluate how the policy's two features mediate inflation's effect on output and employment, we consider two counterfactuals. In the first counterfactual, we evaluate the importance of nominal wage rigidity arising from staggered indexation by considering a scenario where the minimum wage updates every period. In the second counterfactual, we evaluate the importance of anticipation by introducing a cost-push *news* shock to the minimum wage directly. We show that greater nominal wage rigidity amplifies inflation's ability to lower low-skill worker's real wage, while anticipation dampens it by weakening the inflationary shock itself.

5.1 Model Setup

Formally, our economy is composed of n competitive input firms, a continuum of monopolistically competitive intermediate goods producers, a representative final good producer, and a representative household that provides both high and low skill labor. Input firms produce competitively using high and low skill workers and differ in their relative demand for each type of labor. The monopolistically competitive intermediate goods firms produce differentiated output using the n inputs. Then, a competitive final good producer aggregates the intermediate goods. The representative household supplies its labor freely across sectors. Low skill labor is remunerated by the minimum wage, while high skill labor is compensated by a competitive wage rate.

Households and Wage Setting

A representative household derives utility from consumption, C_t , and disutility from labor. It supplies both high and low skill labor, denoted by H_t and L_t , freely across sectors. Their disutility of labor is given by ψ_H and ψ_L respectively. ν is the inverse Frisch elasticity. High skill labor is paid a competitive high-skill wage, denoted W_{Ht} , and low skill labor is paid the minimum wage, denoted W_{Lt} . Following Glover (2016), we incorporate the minimum wage by imposing an upper-bound constraint on low-skill labor. Additionally, households have access to a bond market. Households maximise:

$$\max_{C_t, H_t, L_t, B_{t+1}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma}}{1-\sigma} - \psi_H \frac{H_t^{1+\nu}}{1+\nu} - \psi_L \frac{L_t^{1+\nu}}{1+\nu} \right], \quad (5)$$

subject to:

$$P_t C_t + B_{t+1} = B_t R_{t-1} + W_{Ht} H_t + W_{Lt} L_t + \Gamma_t, \quad (6)$$

$$L_t \leq \bar{L}_t. \quad (7)$$

As in Glover (2016) and Schmitt-Grohé and Uribe (2016), the minimum wage constraint on low-skill labor is exogenous. This implies that low-skill labor is demand-determined. We define the stochastic discount factor as $Q_{t+k|t} = \beta(C_{t+k}/C_t)^{-\sigma}$. Optimization determines high skill labor supply and intertemporal substitution of consumption as follows

$$\frac{W_{Ht}}{P_t} = \psi_H H_t^\nu C_t^\sigma, \quad (8)$$

$$1 = R_t \mathbb{E}_t \left[Q_{t+1|t} \frac{P_t}{P_{t+1}} \right]. \quad (9)$$

Minimum Wage Policy

The minimum wage is indexed to past inflation and updates once every four quarters. The minimum wage rule is given by

$$W_{Lt} = \begin{cases} W_{L,t-1} + \pi_{t-1} + \pi_{t-2} + \pi_{t-3} + \pi_{t-4} & \text{if } t \text{ is the first quarter,} \\ W_{L,t-1} & \text{otherwise.} \end{cases} \quad (10)$$

Firms and Price Setting

Final good producer

A competitive final good firm aggregates intermediate goods $Y_t(j)$ using a CES aggregator. The profit maximisation problem is

$$\max_{Y_t(j)} P_t Y_t - \int_0^1 P_t(j) Y_t(j) dj \quad \text{s.t. } Y_t = \left(\int_0^1 Y_t(j)^{\frac{1}{\lambda^p}} dj \right)^{\lambda^p}, \quad (11)$$

Where P_t is the final good price index, $P_t(j)$ is the price of intermediate good j , and $\lambda^p > 1$ is the elasticity of substitution across intermediate goods. Profit maximisation yields the standard expression for demand of variety j in terms of overall demand for the final good, price of variety j , and the final good price.

$$Y_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{\frac{\lambda^p}{1-\lambda^p}} Y_t. \quad (12)$$

Imposing zero profits, the final price index given by:

$$P_t = \left(\int_0^1 P_t(j)^{\frac{1}{1-\lambda^p}} dj \right)^{1-\lambda^p}. \quad (13)$$

Intermediate goods producers

A continuum of monopolistically competitive intermediate goods producers indexed by $j \in [0, 1]$ use inputs $\{X_{it}(j)\}_{i=1,\dots,n}$ from each input firm to produce a differentiated good $Y_t(j)$ using a constant returns to scale technology. Their profit maximisation problem is given by

$$\max_{\{X_{it}(j)\}_{i=1,\dots,n}} P_t(j)Y_t(j) - \sum_{i=1}^n P_{it}X_{it}(j) \quad \text{s.t.} \quad Y_t(j) = \prod_{i=1}^n X_{it}(j)^{\lambda_i}, \quad (14)$$

Where the elasticity of substitution across inputs is given by λ_i such that $\sum_{i=1}^n \lambda_i = 1$. The cost of each input i , given by P_{it} , dictates its relative demand by each intermediate producer. It is given by

$$\frac{X_{i't}(j)}{X_{it}(j)} = \frac{P_{it}}{P_{i't}}. \quad (15)$$

Intermediate producer j 's absolute demand for input i will depend on overall demand for variety j . The nominal marginal cost $MC^{(n)}$ faced by intermediate producer j will also be equal across intermediate producers and is given by

$$MC^n = \prod_{i=1}^n \lambda_i^{-\lambda_i} P_{it}^{\lambda_i}. \quad (16)$$

Price Setting

We assume that the intermediate goods producers are subject to a Calvo-style friction and can only change their prices infrequently. Each period, a fraction $1 - \xi_p$ of firms can reset their price. Intermediate good producer j sets its price $P_t(j)$ to maximise profits as follows

$$\max_{P_t(j)^*} \sum_{k=0}^{\infty} \xi_p^k \mathbb{E}_t \left\{ Q_{t+k|t} \left(P_t(j)^* Y_{t+k|t}(j) - TC_{t+k|t}^n(Y_{t+k|t}(j)) \right) \right\}, \quad (17)$$

subject to

$$Y_{t+k|t}(j) = \left(\frac{P_t(j)^*}{P_{t+k}} \right)^{\frac{\lambda_p}{1-\lambda_p}} Y_{t+k} \quad \forall k. \quad (18)$$

Where $Q_{t+k|t}$ is the stochastic discount factor for nominal payoffs. And $Y_{t+k|t}(j)$ denotes demand for output in period $t+k$ of firm j that last reset its price in period t . Note that, given our assumptions, all intermediate firms that re-optimize will set the same price.

Input goods producers

Each input firm, indexed by $i \in \{1, \dots, n\}$, produces and input X_{it} competitively using a CES aggregate of high and low skill labor. They maximise profits subject to their production function:

$$\max_{L_{it}, H_{it}} P_{it}X_{it} - W_{Lt}L_{it} - W_{Ht}H_{it} \quad \text{s.t.} \quad X_{it} = \left[\alpha_i \theta L_{it}^{\frac{\eta-1}{\eta}} + (1 - \alpha_i) H_{it}^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \quad (19)$$

η is the elasticity of substitution between labor types, θ governs the efficiency of low skill workers relative to high skill workers and is always less than 1. α_i is the share of low skill workers (per efficiency units) used in production, it is input firm-specific.

The solution to the input firm problem gives demand for each type of labor H_{it} and L_{it} as a function of wages, and the quantity of input produced as follows

$$L_{it} = \left[\frac{(W_{Ht} \theta \alpha_i)^{\eta-1}}{(\theta \alpha_i)^{\eta} W_{Ht}^{\eta-1} + (1 - \alpha_i)^{\eta} W_{Lt}^{\eta-1}} \right]^{\frac{\eta}{\eta-1}} X_{it}, \quad (20)$$

$$H_{it} = \left[\frac{((1 - \alpha_i) W_{Lt})^{\eta-1}}{(\theta \alpha_i)^{\eta} W_{Ht}^{\eta-1} + (1 - \alpha_i)^{\eta} W_{Lt}^{\eta-1}} \right]^{\frac{\eta}{\eta-1}} X_{it}. \quad (21)$$

Finally, this implies that marginal cost for input producers can be expressed as a function of wages as follows

$$MC_{it} = W_{Lt} W_{Ht} \left[(\theta \alpha_i)^{\eta} W_{Ht}^{\eta-1} + (1 - \alpha_i)^{\eta} W_{Lt}^{\eta-1} \right]^{\frac{1}{1-\eta}}. \quad (22)$$

Since input producers are competitive, we have that $P_{it} = MC_{it}$.

Monetary Policy, and Exogenous Processes, and Market Clearing

Monetary Policy

The central bank sets the nominal interest rate $\tilde{R}_t$ according to a standard Taylor rule:

$$\tilde{R}_t = \phi_\pi \hat{\pi}_t + \phi_y \hat{y}_t, \quad (23)$$

where ϕ_π and ϕ_y are the Taylor rule coefficients for inflation and output, respectively.

Exogenous Processes

The economy is driven by a transitory shock to the Phillips curve. It follows the AR(1) process given by:

$$\varepsilon_t^\pi = \rho_\pi \varepsilon_{t-1} + \epsilon_t^\pi, \quad (24)$$

where $\epsilon_t^\pi \sim N(0, \sigma_\pi^2)$.

Market Clearing

Market clearing gives us the final equilibrium conditions that must hold in the model. First, the final goods market clearing condition states that the supply of final good Y_t equals the demand from households C_t

$$Y_t = C_t. \quad (25)$$

Input market clearing requires that the supply of each input X_{it} equal total demand from all the intermediate goods producers.

$$X_{it} = \int_0^1 X_{it}(j) dj \quad \forall i \in \{1, \dots, n\}. \quad (26)$$

Finally, the labor market clears for each labor type. All the supply of high and low skill labor from households equals demand for each of the labor types from the inputs goods producers.

$$L_t = \sum_{i=1}^n L_{it}, \quad H_t = \sum_{i=1}^n H_{it}. \quad (27)$$

Equilibrium

An equilibrium is prices $\{P_t(j), MC_t^n, \{P_{it}\}_{i=1}^n, W_{Lt}, W_{Ht}, R_t\}$ and quantities $\{Y_t, C_t, Y_t(j), \{X_{it}\}_{i=1}^n, \{X_{it}(j)\}_{i=1}^n, H_t, L_t, \{H_{it}\}_{i=1}^n, \{L_{it}\}_{i=1}^n, B_t\}$ such that, given the exogenous processes and government policy, all agents are optimizing and all markets clear.

5.2 Parametrization and Model Solution

To solve the model, we set $N=2$ and log-linearize the equilibrium conditions around the steady-state. The full set of equations is presented in Appendix B. Special attention should be given to the minimum wage indexation rule which is described by the following law of motion:

$$w_t^{\min} = \left(1 - \sum_{k=1}^4 q_{kt} a_k\right) (w_{t-1}^{\min} + \pi_{t-1} + \pi_{t-2} + \pi_{t-3} + \pi_{t-4}) + \left(\sum_{k=1}^4 q_{kt} a_k\right) w_{t-1}^{\min}. \quad (28)$$

where $(1 - a_k)$ is the fraction of workers whose wage adjusts in each quarter k with $k = 1, \dots, 4$, and q_{kt} are dummy variables that take on the value of one when period t is equal to quarter k . As in Olivei and Tenreyro (2007), this implies that wage setting is time-dependent. Therefore, the system is nonlinear and we use the nonlinear solution method of Fuhrer and Bleakley (1996) to solve the model.

Table 2 presents the parametrization of the model. We set the minimum wage to update once a year in the first quarter. Accordingly $1 - a_k$, the probability that the minimum wage resets in quarter k , is equal to 1 if $k = 1$ and equal to 0 if $k \neq 1$. The parameter α_i governs the level of exposure of input firm i to the minimum wage. We set $\alpha_1 < \alpha_2$ such that input firm 1 is more high skill intensive relative to input firm 2. In the linearized conditions, α_i enters the parameter s_L^i which describes the share of firm-level output paid to low skill labor. We set $s_L^1 = 0.1$ and $s_L^2 = 0.9$.

The remaining parameters are standard in the New Keynesian literature. The discount factor β is equal to $1.04^{-\frac{1}{4}}$ which corresponds to an annualized real interest rate of 4%. The intertemporal elasticity of substitution σ is equal to 2, the inverse frisch elasticity ν is equal to 1, the price stickiness parameter ξ_p is equal to 0.35, and the taylor rule coefficients ϕ_π and ϕ_y are equal to 2 and 0.6 respectively.

Table 2: Model Parametrization

Parameter	Description	Value	Source
β	Discount factor	0.99	Standard
σ	Intertemporal elasticity of substitution	2.0	Standard
ν	Inverse Frisch elasticity	1.0	Standard
ϕ_π	Taylor rule inflation coefficient	2.0	Standard
ϕ_y	Taylor rule output coefficient	0.6	Standard
ξ_p	Price stickiness parameter	0.35	Standard
λ	Elasticity of substitution across input goods	0.5	Standard
η	Elasticity of substitution between labor types	0.8	Standard
s_L^1	Low skill labor share in input firm 1	0.0	Calibrated
s_L^2	Low skill labor share in input firm 2	0.05	Calibrated
$1 - a_1$	Probability min wage resets in quarter 1	1	Assumed
$1 - a_2, 1 - a_3, 1 - a_4$	Probability min wage resets in quarter 2, 3, 4	0	Assumed

5.3 Propagation of a cost-push shock

To examine how an inflation-indexed minimum wage like the one in Brazil shapes the propagation of nominal shocks, we start with a baseline scenario where we introduce a cost-push shock and the minimum wage is indexed every four quarters to the accumulated inflation over the past year. Then we consider two extensions designed to highlight the mechanisms through which the policy interacts with the inflation as grease mechanism. In the first extension, we analyze the role of staggered indexation by analysing how a cost-push shock propagates when the minimum wage updates every period. We show that it reinforces the inflation as grease mechanism. In the second extension, we analyze the role of anticipation by introducing a cost-push *news* shock directly to the minimum wage under the same annual indexation policy. We find it dampens the inflation as grease mechanism.

Baseline: Propagation of cost-push shock with annual indexation

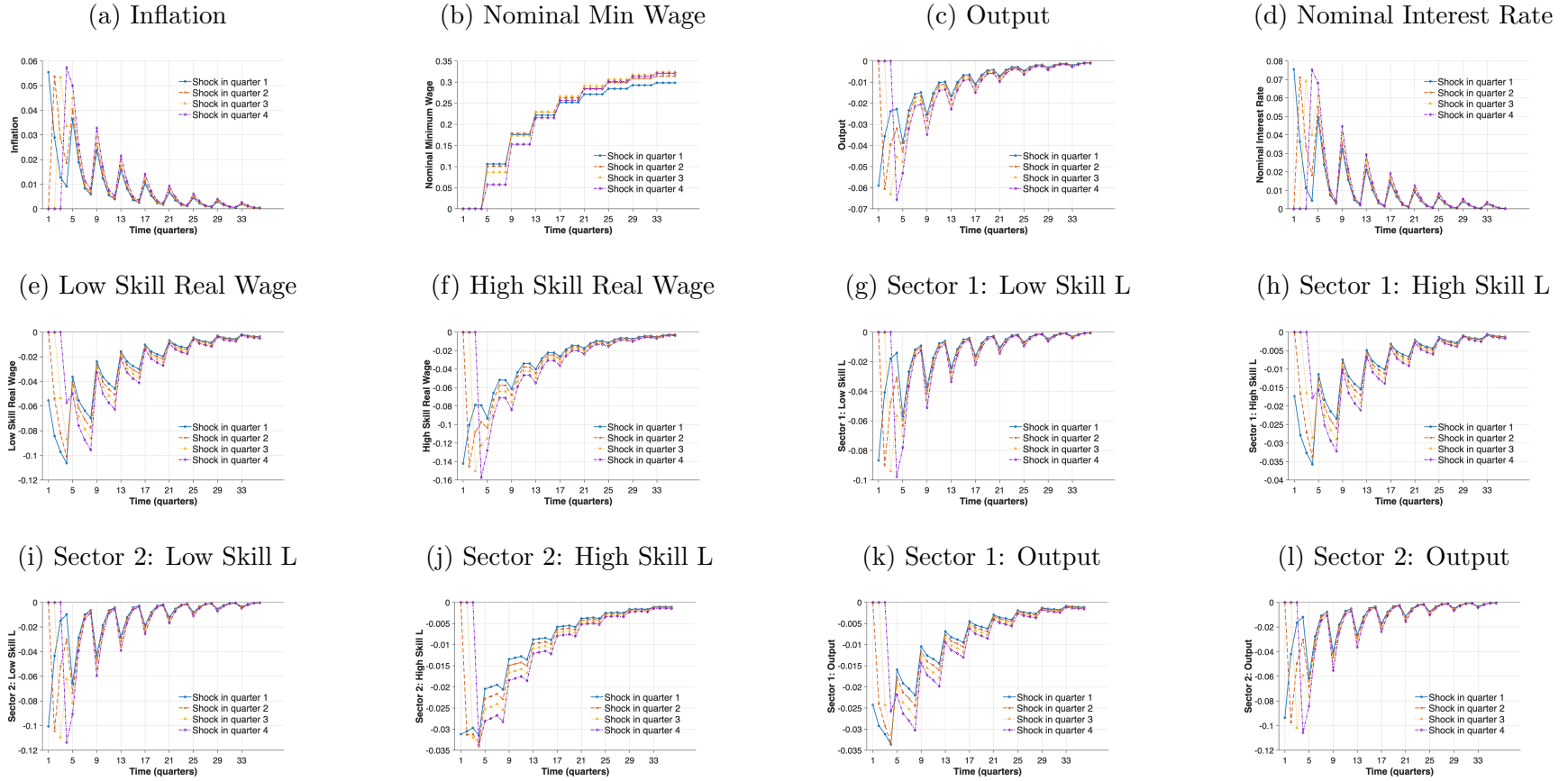
Figure 6 describes how each variable responds to four different 25 basis point aggregate cost-push shocks in each of the first four quarters. In general, the variables respond as expected. Inflation, described in Panel 6a, rises. Monetary policy, described in Panel 6d, reacts by setting a higher nominal interest rate. This causes output, described in Panel 6c, to fall. Then, they return to steady state; inflation and the nominal interest rate fall while output rises. However, their trajectory is interrupted by the first indexation event. The nominal minimum wage, described in Panel 6b, evolves in a step function manner. It rises every first quarter taking into account inflation over the previous four periods, until it eventually stabilizes. With every new indexation event, inflation and the nominal interest rate spike

upward while output spikes downwards, although by less than in response to the initial shock. The economy continues to cycle through this pattern following each subsequent minimum wage increase until the shock dissipates altogether.

In the short run, the effect of the shock is the largest when it takes place in the first quarter (blue line), right after the minimum wage has just been indexed. This is in line with the results in Olivei and Tenreyro (2007) who find that monetary policy shocks have the greatest impact when they take place right after the period of contract renewal. However, we find that in the long run this type of shock has the smallest effect. This can be explained by the interaction between each indexation event, its effect on inflation, and the role of nominal price rigidities. The first minimum wage increase is greatest in response to a first quarter shock because there is more time for inflation to accumulate before the indexation event takes place. However, it also has more time to decline which means that it will rise from a lower base. Thus, in the long run the peaks in inflation are lower and the minimum wage rises by less.

The shock also has important distributional effects. Panels 6e and 6f show that both low and high skill real wages fall, but for different reasons. Since low skill nominal wages are completely rigid, low skill real wages only fall because of the increase in inflation. On the other hand, high skill nominal wages which are completely flexible, are able to fall in response to the output decline. So, while employment falls for both groups due to the decline in aggregate demand, it does so more strongly for low skill workers as firms substitute towards the relatively cheaper high skill labor.

Figure 6: Impulse Response Functions to a Cost-Push Shock When Minimum Wage Indexes Annually



Notes: This figure plots the four impulse responses to a 25 basis point cost-push shock in the first, second, third and fourth quarters separately. Input sector 1 is more high skill intensive while input sector 2 is more low skill intensive.

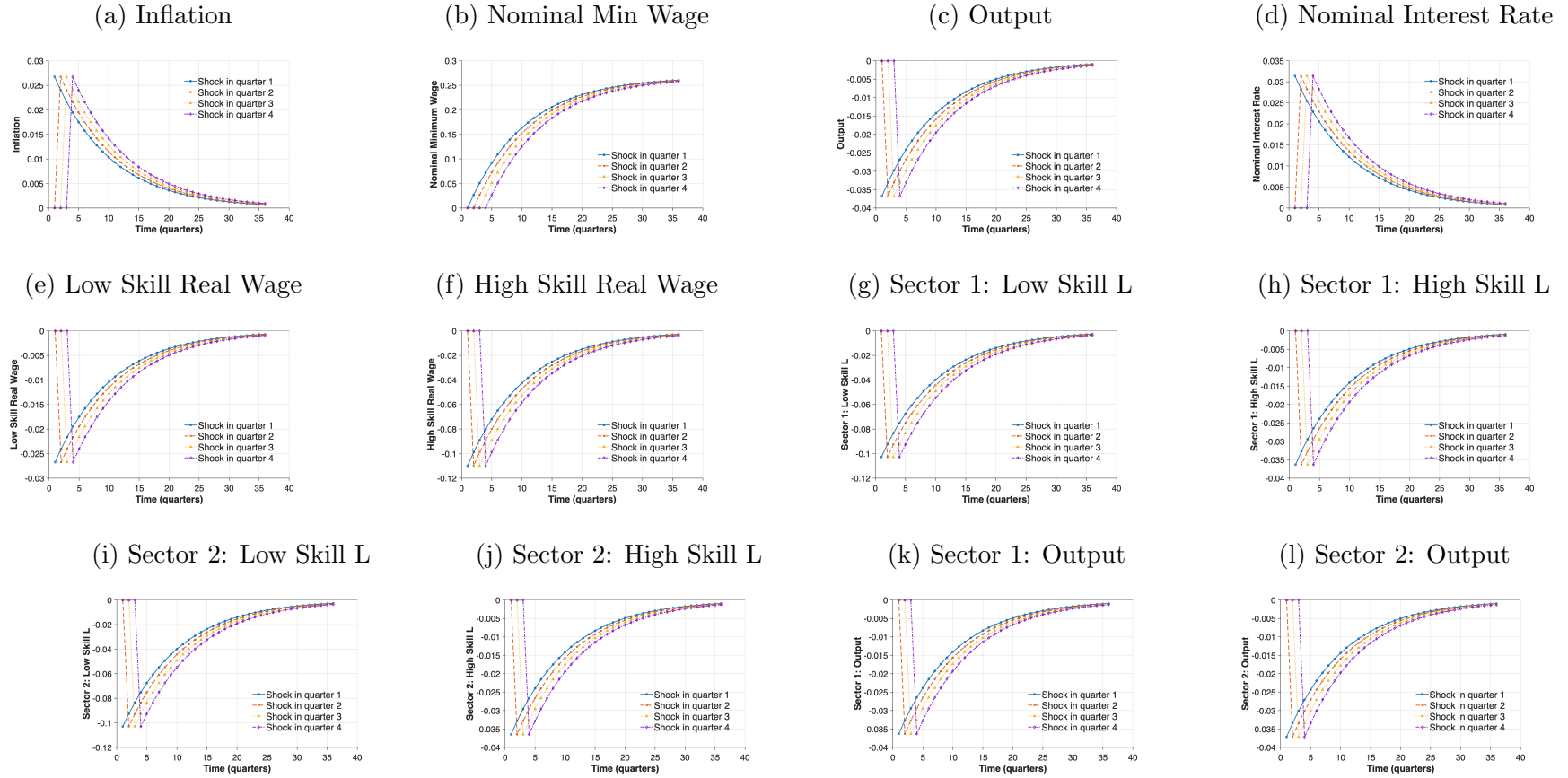
Counterfactual 1: Propagation of cost-push shock with quarterly indexation

Next, we compare how a cost-push shock propagates when the minimum wage updates every quarter. Figure 7 describes the impulse responses to four separate 25 basis point cost-push shock in each of the first four quarters. The direction of each variable is the same as in the baseline case. Inflation rises, monetary policy reacts by raising the nominal interest rate, and output falls. Both the low skill and the high skill real wage fall, although the low skill real wage does so by less.

However, there are two important differences to the baseline case. First, and as expected given the indexation rule, the propagation of the cost-push shock is identical regardless of the quarter in which the shock is introduced. Second, the overall incidence of inflation, and therefore of variation of all the variables, is higher. This is because in the baseline scenario, the rigidity of the nominal minimum wage in between indexation events allows the real low skill wage to fall by more. This boosts labor demand and by extension output - exactly highlighting the inflation as grease mechanism. Ultimately this force counteracts the output decline induced by the minimum wage increase.

Since the minimum wage updates every period, there is no role for anticipation.

Figure 7: Impulse Response Functions to a Cost-Push Shock When Minimum Wage Indexes Every Period



Notes: This figure plots the four impulse responses to a 25 basis point cost-push shock in the first, second, third and fourth quarters separately. Input sector 1 is more high skill intensive while input sector 2 is more low skill intensive.

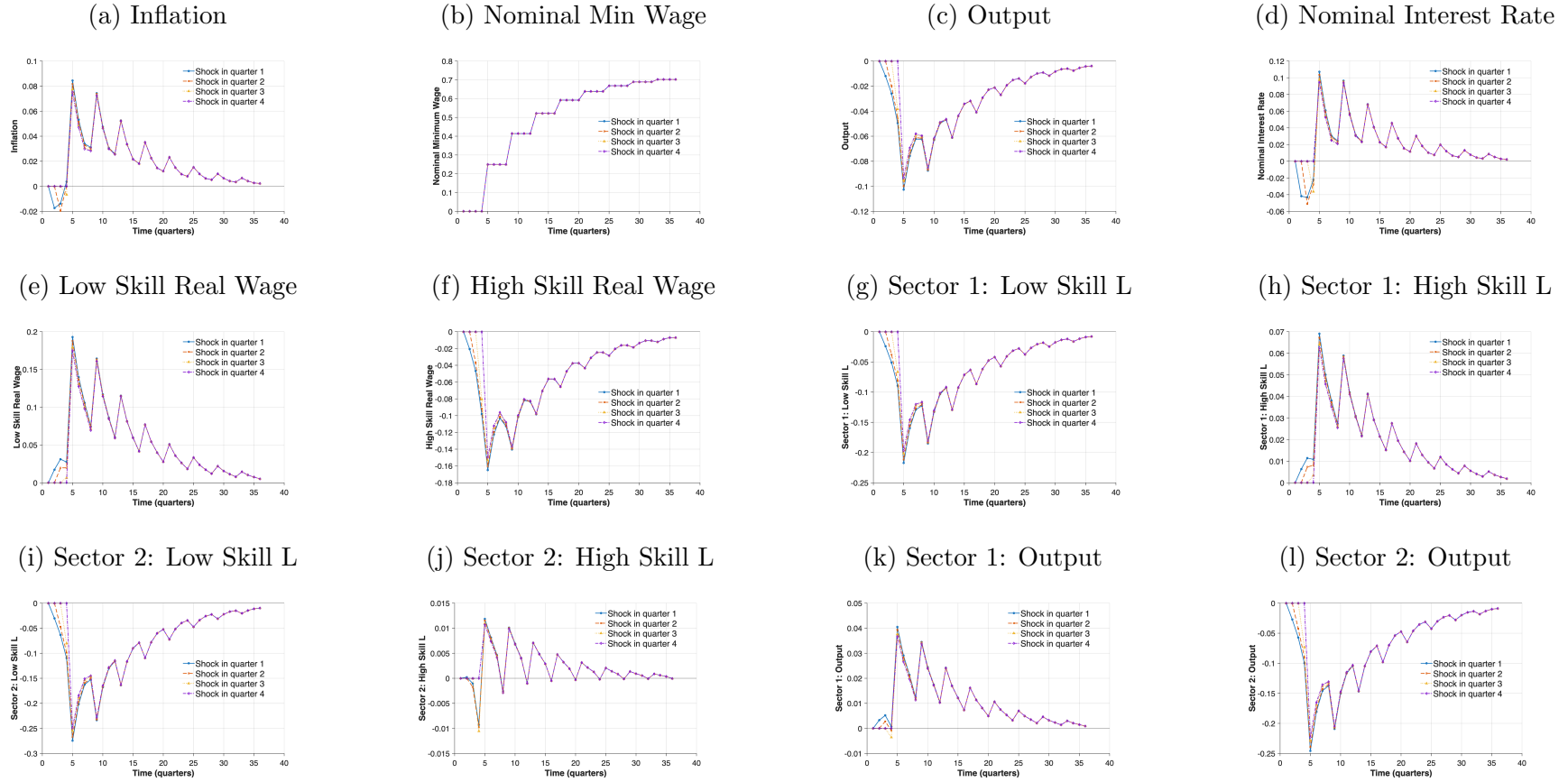
Counterfactual 2: Propagation of cost-push *news* shock with annual indexation

Finally, to isolate the effect of a rising minimum wage from the contemporaneous effect of the shock on inflation we replace the cost-push shock to inflation with a cost-push *news* shock directly to the minimum wage under the same annual indexation policy. The results are described in Figure 8. As before, the shocks are introduced in each of the four quarters, but they only directly affect the minimum wage which updates in the first quarter of the following year. To isolate the effect even further, the minimum wage rule does not incorporate accumulated inflation. For this reason, as described in Panel 8b, the minimum wage increases by the same amount regardless of the quarter in which the shock is introduced. Thus, the behavior of the other variables before the minimum wage increase is a result of its anticipation.

First, output falls in anticipation of the minimum wage increase because of intertemporal substitution. The decline in aggregate demand lowers inflation as well. Monetary policy reacts by setting a lower interest rate, but not by enough to reverse the effects. After the initial decline, inflation begins to rise in anticipation of the minimum wage increase because of precautionary pricing. After the first minimum wage increase, the patterns are similar to those in the baseline scenario in response to a cost-push shock. This is because each increase in the nominal minimum wage introduces a marginal cost increase.

The distributional effects are also different to the baseline scenario and are indicative of anticipation working against the inflation as grease mechanism. In the short run, the low skill real wage rises due to the decline in inflation, while the high skill real wage still declines. This puts further downward pressure on output despite the decline in the nominal interest rate. Low skill employment falls by much more compared to the baseline scenario.

Figure 8: Impulse Response Functions to a Cost-Push News Shock When Minimum Wage Indexes Annually



Notes: This figure plots the four impulse responses to a 25 basis point cost-push news shock in the first, second, third and fourth quarters separately. Input sector 1 is more high skill intensive while input sector 2 is more low skill intensive.

6 Conclusion

Overall, we have shown that an inflation-indexed minimum wage interferes with how nominal shocks propagate. Using Brazil as a case study, we establish two empirical results. First, that the policy creates asymmetric upward nominal wage rigidity across exposed and unexposed workers. Second, that firms anticipate the minimum wage increase. Using a New Keynesian model augmented with two worker types and an inflation-indexed minimum wage, we show that the two features of the policy highlighted in the empirical findings have countervailing effects on inflation’s ability to lower the real wage. On the one hand, staggered indexation introduces nominal wage rigidity which amplifies inflation’s ability to lower the real wage and therefore “grease the wheels” of the labor market. On the other hand, anticipation can undo the effects of a nominal shock by lowering output and therefore the strength of the inflation shock itself through intertemporal substitution.

Our findings support the view that policymakers must account for how policies impact agent’s expectations when designing them. In addition, we show how a policy designed to help low-wage workers can have unintended consequences through previously understudied channels, such as by inducing greater nominal wage rigidity. Ultimately, when evaluating how a nominal shock propagates, a deep understanding of the institutional context is key.

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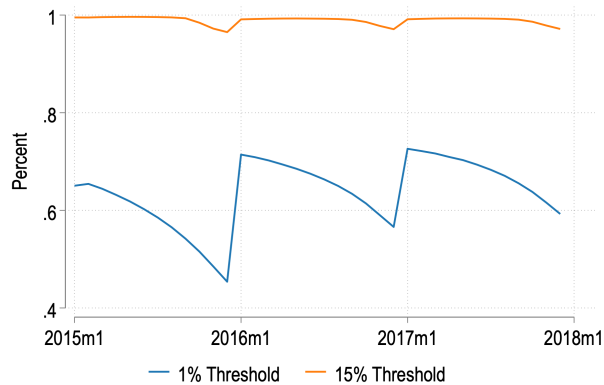
A Appendix: Data

A.1 Measurement Error in Earnings

In this section we describe how we restrict the analysis sample to address some inconsistencies in the RAIS dataset. Specifically, a worker's earnings are recorded in two different ways, and they do not always match. First, starting in 2015 each firm reports the earnings of worker l in year t in each month m , y_{mt}^l . Second, firms directly report a worker's annual average monthly earnings $\hat{y}_t^l$.

It follows that the annual average of a worker's monthly earnings, which we calculate and call $\hat{y}_t^{l,pred}$, should coincide with their directly reported annual average monthly earnings $\hat{y}_t^l$. This is not always the case in the data, which suggests there is measurement error in the earnings variable.

Figure A.1: Percent of verified earnings over time



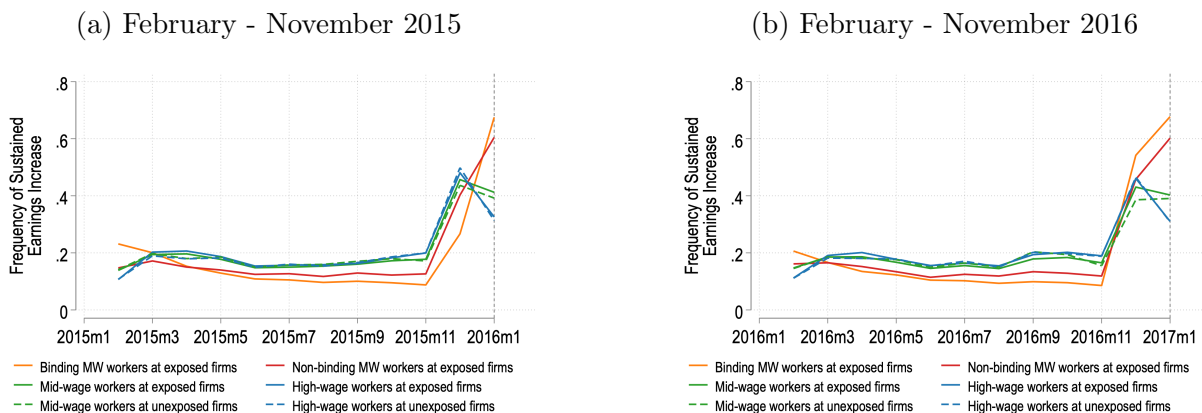
To address this discrepancy, we restrict the main analysis sample to observations where the predicted annual average monthly earnings $\hat{y}_t^{l,pred}$ lies within 1% of the directly reported annual average monthly earnings. The blue line of figure A.1 describes the percent of observations which satisfy this restriction over time. It shows that the percent of verified earnings oscillates between 70% and 50%. Moreover, there is a striking seasonal pattern. It jumps in January, and then falls steadily until December. Within a year, the percent of verified

earnings can vary across months because of workers' entry into and exit out of the formal labor market.

15% Threshold Sample

In this section, we replicate the main results using an alternative sample where we loosen the criteria to include all observations where the directly reported annual average monthly earnings lies within 15% of the predicted annual average monthly earnings. The orange line of figure A.1 describes the percent of observations which satisfy this restriction over time. In this case, the percent of verified earnings lies very close to 100% throughout the whole sample.

Figure A.2: Upward Nominal Wage Rigidity using Alternative Sample

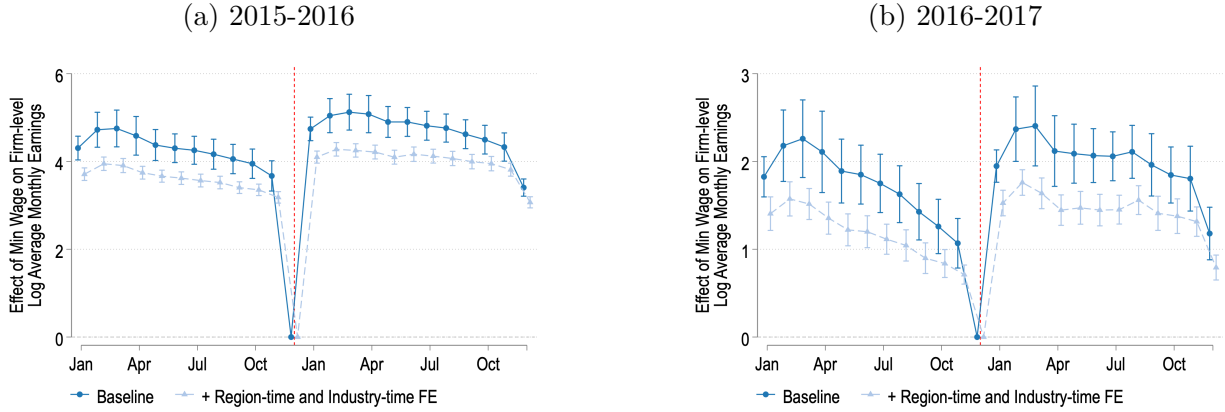


Notes: This figure presents non-parametric evidence on the frequency of sustained earnings increases at the worker-level using the alternative sample described in section A.1. The frequency of sustained earnings increases is defined as the share of workers with a month-to-month earnings increase that is at least maintained the following month. Workers are classified when they enter the sample. Binding MW workers are those that earn exactly the current minimum wage. Non-binding MW workers are those that earn more than the current min wage but less than the following year's minimum wage. Mid-wage workers are those that earn more than the following year's minimum wage but less than two multiples of it. High-wage workers are those that earn more than two multiples of the following year's minimum wage.

Figure A.2 replicates figure 1 using this alternative sample. The results are qualitatively similar to those presented in the main text. In both years, we observe that minimum wage workers experience fewer month-to-month earnings increases relative to other workers prior to indexation events. This confirms the presence of upward nominal wage rigidity that correlates with the minimum wage indexation policy.

Figure A.3 replicates figure 4 using this alternative sample. Although the patterns are similar, the conclusions are less stark. Panel A.3b supports the statement that the average earnings workers at firms that will be more exposed to the minimum wage increase are

Figure A.3: Earnings at the Firm level using Alternative Sample



Notes: The figure plots the event study coefficients β_m^{Firm} from estimating equation 4 using firm-level average earnings as the outcome and using the alternative sample. In dark blue circles we report the coefficients from the baseline regression, and in light blue triangles we report the coefficients from the regression controlling for firm's region and sector. The regressions are weighted by each firm's average employment during the sample period.

growing slower relative to less exposed firms prior to indexation events. However, this is not so evident in panel A.3a.

Distribution of Measurement Error

To further our understanding of whether the measurement error we have identified matters for our results, we analyze its prevalence throughout the distribution of workers.

Figure A.4 compares, for workers whose earnings are unverified, the distribution of their directly reported annual average monthly earnings $\hat{y}_t^l$ in blue, with the distribution of their predicted annual average monthly earnings $\hat{y}_t^{l,pred}$ in red. In all three years, the distribution of $\hat{y}_t^{l,pred}$ exhibits a long left tail. This reveals that when monthly earnings are mis-reported, they are consistently recorded to be “too low”.

Figure A.4: Distribution of Measurement Error in Earnings (Identified Data)

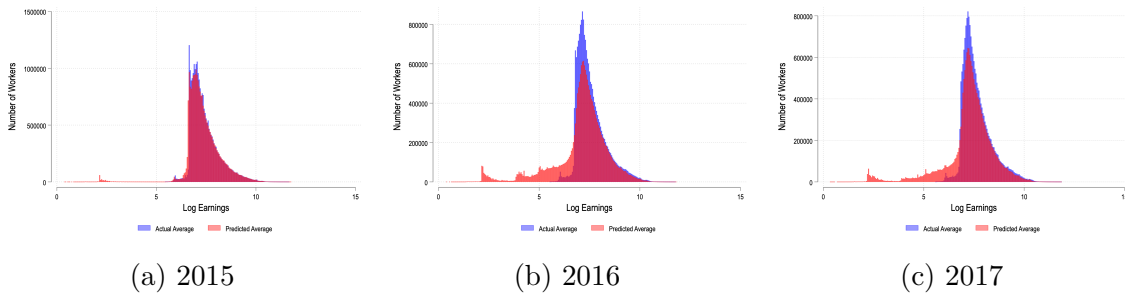
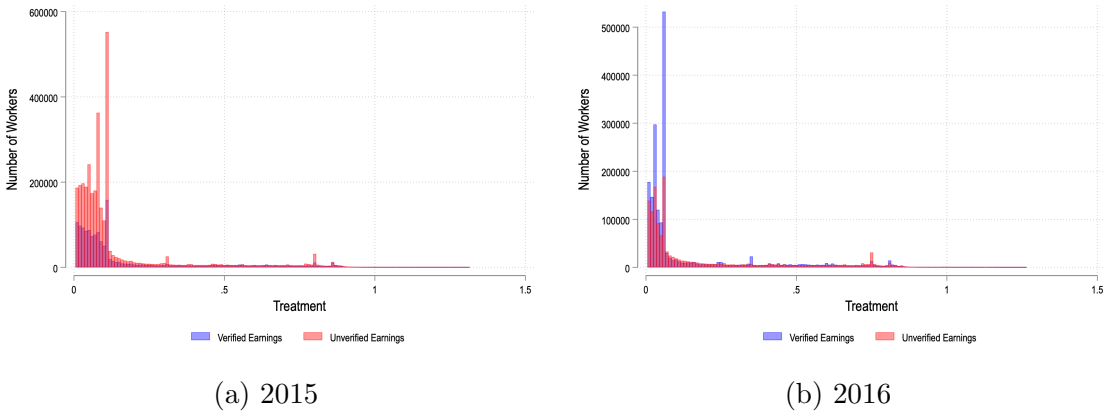


Figure A.5 compares the distribution of treatment for workers with verified earnings in blue, with the distribution of treatment for workers with unverified earnings in red. Treatment is calculated as the predicted earnings increase due to the minimum wage: $\ln(mw_{t+1}) - \ln(\hat{y}_t^l)$. The figure restricts to workers whose treatment value is greater than zero. Although different in magnitude, the value of treatment across the two sets of workers is distributed similarly. This suggests that using the directly reported annual average monthly earnings $\hat{y}_t^l$ to calculate treatment ameliorates any concerns that it may be affected by measurement error in the earnings variable.

Figure A.5: Distribution of Measurement Error in Treatment (Identified Data)



A.2 Summary Statistics by Earnings Percentiles

Table A.1 reports the summary statistics by earnings percentiles for the sample 2016-2017. It is similar to Table 1, which reports the same statistics for the year 2015.

A.3 Non-parametric frequency of sustained earnings increases

Here, we repeat the exercise in figure 1 but classify firms into three exposure groups instead of two. These are unexposed, low exposure and high exposure firms. Low and high exposure firms lie above and below the median level of exposure respectively.

While the results are qualitatively similar to those presented in the main text, figure A.6 does present two challenges. First, minimum wage workers at low exposure firms (orange line, dots) behave more similarly to non-minimum wage workers at unexposed firms (green and blue lines, dash). This suggests that exposed workers at low exposure firms are less affected by upward nominal wage rigidity. However, the share of minimum wage workers at low exposure firms is very small.

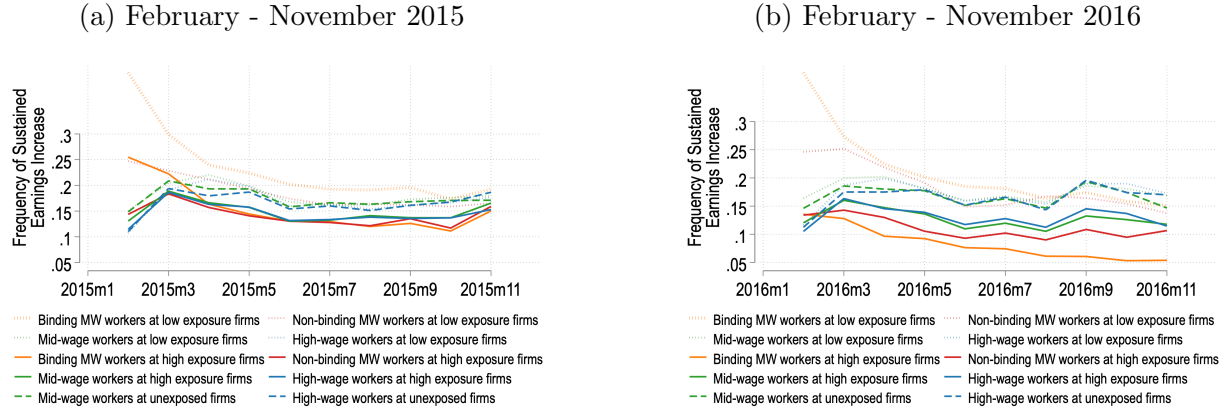
Table A.1: Summary Statistics by Earnings Percentiles (2016)

	$\hat{y}_t < mw^{t+1}$	$mw^{t+1} < \hat{y}_t < 2 \cdot mw^{t+1}$	$2 \cdot mw^{t+1} < \hat{y}_t$
Observations	1035718	6928745	3252501
Female	0.42	0.40	0.24
Age	33.41	34.51	38.15
Education			
Less than High School	0.49	0.40	0.26
High School	0.47	0.53	0.45
More than High School	0.04	0.07	0.28
Establishment Size			
Small (<20)	0.26	0.26	0.18
Medium (20–250)	0.32	0.36	0.40
Large (>250)	0.42	0.38	0.42
Regions			
North	0.08	0.05	0.04
Northeast	0.41	0.18	0.10
Southeast	0.36	0.50	0.58
South	0.07	0.18	0.20
Centerwest	0.09	0.09	0.08
Sectors			
Agr. and Mining	0.22	0.09	0.06
Manuf. and Constr.	0.24	0.27	0.38
Services	0.53	0.63	0.55

Notes: This table presents summary statistics for workers in three groups defined by their average monthly earnings in 2016 relative to the upcoming minimum wage increase in January 2017. The first column includes workers who earn less than the 2017 minimum wage, the second column groups workers who earn more than that but less than two multiples of it, and the third column includes workers who earn more than two multiples of it.

Second, non-minimum wage workers at high exposure firms (green and blue lines, solid lines) behave more similarly to minimum wage workers. However, while it is true that their frequency of earnings increases lies below that of non-minimum wage workers at unexposed firms, it is still well above that of minimum wage workers at the same type of firm.

Figure A.6: Non-parametric Evidence on Upward Nominal Wage Rigidity



Notes: This figure presents non-parametric evidence on the frequency of sustained earnings increases at the worker-level. The frequency of sustained earnings increases is defined as the share of workers with a month-to-month earnings increase that is at least maintained the following month. Workers are classified when they enter the sample. Binding MW workers are those that earn exactly the current minimum wage. Non-binding MW workers are those that earn more than the current min wage but less than the following year's minimum wage. Mid-wage workers are those that earn more than the following year's minimum wage but less than two multiples of it. High-wage workers are those that earn more than two multiples of the following year's minimum wage. Firms are classified into unexposed, low exposure, and high exposure firms. Unexposed firms have no workers earning less than the following year's minimum wage. Low exposure firms are those whose exposure share lies below the median exposure level, while high exposure firms are those that lie above the median exposure.

B Appendix: Model with 2 Input Firms

The system of log-linearized equations is giving by:

Household Block

$$\text{Intertemporal Euler equation :} \quad y_t = y_{t+1} - \frac{1}{\sigma} (r_t - \pi_{t+1})$$

$$\text{High-skill wage determination :} \quad w_t^H = \nu h_t + \sigma y_t$$

Firm Block

$$\begin{aligned}
\text{New Keynesian Phillips Curve :} \quad & \pi_t = \frac{(1 - \xi_p \beta)(1 - \xi_p)}{\xi_p} mc_t + \beta \pi_{t+1} + \varepsilon_t^\pi \\
\text{Cost minimization condition :} \quad & x_t^2 - x_t^1 = q_t^1 - q_t^2 \\
\text{Final goods production function :} \quad & y_t = \lambda x_t^1 + (1 - \lambda) x_t^2 \\
\text{Marginal cost of final goods producers :} \quad & mc_t = \lambda q_t^1 + (1 - \lambda) q_t^2 \\
\text{Sector 1, low-skill labor demand :} \quad & w_t^L = q_t^1 + \frac{1}{\eta} (x_t^1 - l_t^1) \\
\text{Sector 1, high-skill labor demand :} \quad & w_t^H = q_t^1 + \frac{1}{\eta} (x_t^1 - h_t^1) \\
\text{Sector 2, low-skill labor demand :} \quad & w_t^L = q_t^2 + \frac{1}{\eta} (x_t^2 - l_t^2) \\
\text{Sector 2, high-skill labor demand :} \quad & w_t^H = q_t^2 + \frac{1}{\eta} (x_t^2 - h_t^2) \\
\text{Sector 1 - marginal cost :} \quad & q_t^1 = s_{L,1} w_t^L + (1 - s_{L,1}) w_t^H \\
\text{Sector 2 - marginal cost :} \quad & q_t^2 = s_{L,2} w_t^L + (1 - s_{L,2}) w_t^H
\end{aligned}$$

Market Clearing Conditions

$$\begin{aligned}
\text{Goods market clearing condition :} \quad & y_t = c_t \\
\text{High-skill labor market :} \quad & h_t = H_1 h_{1,t} + H_2 h_{2,t}
\end{aligned}$$

Policy block and Definitions

$$\begin{aligned}
\text{Monetary policy :} \quad & r_t = \phi_\pi \pi_t + \phi_y y_t \\
\text{Inflation :} \quad & \pi_t = p_t - p_{t-1} \\
\text{Low skill real wage :} \quad & w_t^L = w_t^{\min} - p_t \\
\text{Law of motion min wage :} \quad & w_t^{\min} = \left(1 - \sum_{k=1}^4 q_{kt} * a_k \right) * (w_{t-1}^{\min} + \pi_{t-1} + \pi_{t-2} + \pi_{t-3} + \pi_{t-4}) \\
& + \left(\sum_{k=1}^4 q_{kt} * a_k \right) * w_{t-1}^{\min}
\end{aligned}$$

Note that the parameters s_L^1 and s_L^2 govern the share of low skill labor in each input firm. They represent the share of output of firm i that is attributable to low skill labor. They are

given by:

$$s_L^i = \frac{(\theta\alpha_i)L_i^{\frac{\eta-1}{\eta}}}{(\theta\alpha_i)L_i^{\frac{\eta-1}{\eta}} + (1-\alpha_i)H_i^{\frac{\eta-1}{\eta}}} \quad (29)$$